

RESEARCH ARTICLE

WILEY

A simple Eulerian–Lagrangian weakly compressible smoothed particle hydrodynamics method for fluid flow and heat transfer

Hee Sang Yoo  | Young Beom Jo | Jin Woo Kim | Eung Soo Kim | Tae Soo Choi

Department of Nuclear Engineering, Seoul National University, Seoul, South Korea

Correspondence

Eung Soo Kim, Department of Nuclear Engineering, Seoul National University, 1 Gwanak-ro, Gwanak-gu, Seoul, South Korea.

Email: kes7741@snu.ac.kr

Funding information

National Research Foundation of Korea, Grant/Award Numbers: 2019R1F1A1060159, 2021M2D2A1A03046881

Abstract

In this article, we propose a simple, consistent, and robust Eulerian–Lagrangian weakly compressible smoothed particle hydrodynamics (EL-WCSPH) for the simulation of hydrodynamics and convection heat transfer problems. First, the basic governing equation is formulated using the ALE-SPH framework. The universal formulation is similar to the standard WCSPH, but the degree of Eulerian and Lagrangian schemes can be controlled by inserting a pre-defined arbitrary parameter $\alpha \in [0, 1]$. To alleviate potential numerical errors and inconsistency, particle mass correction is proposed along with particle disorder correction. Then, an algorithm is introduced, in which only a slight modification is necessary compared to the standard WCSPH. Finally, extensive benchmark simulations are conducted to validate the proposed method. As a result, the proposed method is found to perform well in hydrodynamics problems with various flow conditions (pipe flow, lid-driven problem, Taylor–Green vortex, vortex shedding problem, and water sloshing) and thermal-hydraulics problems (natural convection in a square cavity).

KEYWORDS

GPU, mixed Eulerian–Lagrangian, particle mass correction, WCSPH

1 | INTRODUCTION

Smoothed particle hydrodynamics (SPH) is a Lagrangian-based meshfree numerical method that traces discretized particles (or parcels) by solving governing equations. This method was first introduced by Lucy¹ and Gingold and Monaghan² for solving astrophysics problems, SPH has progressed significantly and has been applied in many other science and engineering fields.^{3–7} Owing to its Lagrangian nature, SPH offers intrinsic advantages in handling free surface or multi-phase flows with large deformations without requiring additional numerical treatment for interface tracking.

In SPH, there are two approaches to handle liquids. One is based on the complete incompressible assumption and is called incompressible SPH (ISPH). In this approach, the momentum equation is solved in two steps, and the pressure field is calculated by solving the pressure Poisson equation (PPE). The other approach originates from the early form of SPH introduced by Monaghan,⁸ and it is called weakly compressible SPH (WCSPH). In this approach, a slight compression of liquid is allowed, and pressure is calculated using an equation of state (EOS). Both approaches have their own advantages and disadvantages. Generally, ISPH can provide superior pressure field predictions without high frequency noise and manage longer time-step sizes. However, with this scheme, one is required to solve a rather large sparse matrix of PPEs

and perform explicit detection of free surfaces. On the other hand, WCSPH is easily implementable and parallelizable given its straightforward scheme. Historically, this simplicity has encouraged researchers from various fields to apply this method to their domains of science and engineering. However, high-frequency numerical pressure noise and oscillation are frequently encountered in WCSPH. To alleviate these issues, several methods such as δ -SPH^{9,10} and Riemann-based SPH^{11,12} have been introduced, and their effectiveness has been demonstrated.

In recent SPH studies, many researchers have explored Arbitrary Lagrangian Eulerian (ALE) formulations to utilize both Eulerian and Lagrangian natures.^{12–18} The ALE formalism was first proposed in the finite element (FE) framework to combine the advantages of the Eulerian and Lagrangian frameworks; subsequently, it was first introduced in the SPH context by Vila.¹³ In the ALE–SPH scheme, motion of the reference frame is defined by an arbitrary transport velocity ($\mathbf{u}_0$) that differs from the physical velocity ($\mathbf{u}$) of the particle. Therefore, as $\mathbf{u}_0$ approaches zero, the nature of the formulation transitions to pure Eulerian; as $\mathbf{u}_0$ approaches the physical velocity ($\mathbf{u}$), the nature of the formulation transitions to pure Lagrangian; if $\mathbf{u}_0$ in between, the formulation is a combination of both schemes.

According to Vila,¹³ the optimal choice of an appropriate arbitrary transport velocity is critical to increase the stability and robustness of the scheme without degrading its consistency. Vila¹³ implemented the SPH–ALE formulation by using the approximate Riemann solver. The arbitrary transport velocity was determined following the smoothed velocity field, as in the extended SPH (XSPH) method. Vila¹³ expanded his research path, relying on the SPH–ALE combination and a finite volume-stabilizing low-Mach scheme, the so-called γ -SPH-ALE method.¹⁴ The proposed scheme exhibited increased stability and accuracy in high-velocity impact (HVI) modeling¹⁵ and on various hydrodynamics problems.¹⁴ Oger et al.¹² worked with SPH-ALE by using the approximate Riemann solver. They manipulated the transport velocity field according to the distribution of particles, similarly to the particle shifting technique (PST). A uniform particle structure and good improvement in stability were obtained by straightforward inclusion of the shifting velocity into the governing equations as a quasi-Lagrangian mode. Recently, Antuono et al.¹⁶ implemented the SPH-ALE method by using the δ -SPH scheme, analogously to Oger et al.,¹² and achieved outstanding performance on various hydrodynamics problems, including violent impact behavior. Lind and Stansby¹⁷ and Fourtakas et al.¹⁸ adopted the SPH–ALE approach in the ISPH context. They initially determined the arbitrary transport velocity by separating the Eulerian and Lagrangian regions; the cohabitation of both schemes provided the benefits of both schemes. The Eulerian particles were used in the boundary or interior to improve accuracy with a high-order kernel, and the Lagrangian particles were placed near the interface. The new scheme yielded superior results when applied to internal flows and free surface flows.

In this study, we propose a very simple but robust method to acquire ALE capability for the standard WCSPH through slight modification of its formulation. Improved stability and accuracy on the prediction near the wall and force coefficients are achieved by cohabitating of Eulerian and Lagrangian region, which is named as Eulerian–Lagrangian WCSPH (EL-WCSPH). Furthermore, the proposed scheme has been extended to deal with convection heat transfer. One of the key differences between the standard WCSPH and the proposed method is consideration of the mass variations that might occur under the ALE-based governing equations. Oger et al.¹² solved a volume flux equation derived from the Reynolds transport theorem to take into account the variation of volume. Antuono et al.¹⁶ reshaped the volume flux equation and calculated the variation of mass. In this study, the variation of mass and volume are consistently and straightforwardly included by using a simple mass correction method based on the basic principle of SPH approximation. Due to its simplicity, our method can handle not only incompressible flow but also compressible flow with low Mach number correctly. Also, its extension to multi-resolution approach is logical. To prevent the tensile instability of particles and penetration between regions, the particle disorder correction algorithm is adopted. The accuracy and stability of the proposed method are guaranteed under the complement between the particle mass correction and particle disorder correction.

The remainder of this article is organized as follows. In Section 2, the basic concept of SPH and general WCSPH governing equations are briefly introduced. In Section 3, we present the simple EL-WCSPH scheme with our main idea of novel mass correction and particle disorder correction. The 1D shock tube problem is performed and discussed to validate our main idea. In Section 4, several steady (pipe flow, lid-driven problem), and transient (Taylor–Green vortex, vortex shedding problem, and water sloshing) hydrodynamic problems are performed in order to demonstrate the accuracy and convergence of the proposed method, furthermore, the feasibility of the EL-WCSPH method is evaluated through complicated hydrodynamic problem (vortex shedding including free surface) and convective heat transfer (natural convection in a square cavity). Finally, Section 5 briefly summarizes our results and concludes the article with some further discussions.

2 | ALE-SPH DISCRETIZATION

2.1 | SPH approximation

In the basic SPH formulation, a function $f(\mathbf{r})$ at a point $\mathbf{r}$ is approximated as the integral interpolant over a domain of interest Ω by using a smoothing kernel function $W(\mathbf{r} - \mathbf{r}', h)$ with a smoothing length h .

$$f(\mathbf{r}) = \int_{\Omega} f(\mathbf{r}') W(\mathbf{r} - \mathbf{r}', h) d\mathbf{r}' \quad (1)$$

In the discretized system, the fluid system is represented by particles, and the function $f(\mathbf{r})$ is mathematically expressed as follows:

$$f(\mathbf{r}_i) = \sum_j f(\mathbf{r}_j) W_{ij}(\mathbf{r}_{ij}, h) V_j = \sum_j f(\mathbf{r}_j) W_{ij}(\mathbf{r}_{ij}, h) \frac{m_j}{\rho_j}, \quad (2)$$

where $f(\mathbf{r}_i)$ is a scalar or vector function related to the position of the i th particle; subscript j denotes the neighboring particles; and V_j , m_j , and ρ_j are the volume element, mass, and density of the j th particle, respectively. The $\mathbf{r}_{ij}$ is the distance between the i th particle and j th particle. $W_{ij}(\mathbf{r}_{ij}, h) = W_{ij}(|\mathbf{r}_i - \mathbf{r}_j|, h)$ is a kernel function that determines the weighting factor between the particles i and j , and hereinafter, it is denoted by W_{ij} .

In this study, the Wendland sixth kernel is used for overall calculations.¹⁹ A smoothing length is chosen to be $h = 4.5dx$ for the compromise between the numerical stability and computational load, where dx is the initial particle spacing.

$$W_{ij} = W_{ij}(|\mathbf{r}_{ij}|, h) = \begin{cases} \frac{78}{7\pi} \left(1 - \left(\frac{|\mathbf{r}_{ij}|}{h}\right)\right)^8 \left(1 + 8\left(\frac{|\mathbf{r}_{ij}|}{h}\right) + 25\left(\frac{|\mathbf{r}_{ij}|}{h}\right)^2 + 32\left(\frac{|\mathbf{r}_{ij}|}{h}\right)^3\right) & 0 \leq |\mathbf{r}_{ij}| < h \\ 0 & |\mathbf{r}_{ij}| \geq h \end{cases} \quad (3)$$

Similarly, the gradient and divergence of a function can be approximated as follows:

$$\nabla A(\mathbf{r}_i) = \sum_j [A(\mathbf{r}_j) - A(\mathbf{r}_i)] \nabla_i W_{ij} \frac{m_j}{\rho_j}, \quad (4)$$

$$\nabla \cdot \mathbf{a}(\mathbf{r}_i) = \sum_j [\mathbf{a}(\mathbf{r}_j) - \mathbf{a}(\mathbf{r}_i)] \cdot \nabla_i W_{ij} \frac{m_j}{\rho_j}, \quad (5)$$

where A and $\mathbf{a}$ are the scalar and vector functions, respectively, ∇_i is the spatial gradient with respect to the i th particle. Given that many review papers have summarized and explained the basic concepts of SPH and kernel summations of various mathematical formulations, the present paper does not cover them in detail. More information can be found in the literature.²⁰

2.2 | ALE governing equations

In the pure Lagrangian framework, the basic governing equations of fluid flow and heat transfer are as follows under the assumption of the constant thermo-physical properties:

$$\frac{D\rho}{Dt} = -\rho \nabla \cdot \mathbf{u}, \quad (6)$$

$$\frac{D\mathbf{u}}{Dt} = -\frac{\nabla P}{\rho} + \frac{\nabla \tau^v}{\rho} + \mathbf{f}_b, \quad (7)$$

$$\frac{DT}{Dt} = -\frac{k}{c_p} \nabla^2 T, \quad (8)$$

$$P = F(\rho), \quad (9)$$

where ρ is the density, $\mathbf{u}$ is the physical velocity, P is pressure, $\boldsymbol{\tau}^v$ is the viscous stress tensor. $\mathbf{f}_b$ represents the body force per unit mass, which is equal to gravity(= $\mathbf{g}$) in the hydrodynamic problems or acceleration due to buoyant force(= $\beta \mathbf{g}(T - T_0)$) with the convective heat transfer problems. k , c_p , β , T , and T_0 are heat conductivity, specific heat, thermal expansion coefficient, temperature, and reference temperature. The function F is an equation of state that correlates density and pressure. When the fluid is assumed to be weakly compressible, pressure is commonly estimated using the Tait equation⁸:

$$P = F(\rho) = \frac{c_0^2 \rho_0}{\gamma} \left[\left(\frac{\rho}{\rho_0} \right)^\gamma - 1 \right], \quad (10)$$

where ρ_0 is the reference density of the fluid, and c_0 is the speed of sound. γ is the polytropic constant, which determines the sensitivity of pressure to compressibility, and typically, its value is 7.0 for liquids.

The above governing equations are now rewritten based on Eulerian framework and extended to the ALE framework. The material derivative (Df/Dt) of a general fluid variable f is defined as follows:

$$\frac{Df}{Dt} := \frac{\partial f}{\partial t} + \mathbf{u} \cdot \nabla f. \quad (11)$$

Using this definition, Equations (6)–(9) can be rewritten as follows in the Eulerian framework.

$$\frac{\partial \rho}{\partial t} + \mathbf{u} \cdot \nabla \rho = -\rho \nabla \cdot \mathbf{u} \quad (12)$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{\nabla P}{\rho} + \frac{\nabla \boldsymbol{\tau}^v}{\rho} + \mathbf{f}_b \quad (13)$$

$$\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T = -\frac{k}{c_p} \nabla^2 T \quad (14)$$

$$P = F(\rho) \quad (15)$$

Similar to Lind and Stansby,¹⁷ we define the ALE derivative ($\frac{df}{dt}$) with respect to the arbitrary transport velocity ($\mathbf{u}_0$) as follows.

$$\frac{df}{dt} := \frac{\partial f}{\partial t} + \mathbf{u}_0 \cdot \nabla f = \frac{\partial f}{\partial t} + \alpha \mathbf{u} \cdot \nabla f. \quad (16)$$

If the arbitrary transport velocity ($\mathbf{u}_0$) has the same direction but a different magnitude compared to the physical velocity ($\mathbf{u}$) for convenience, $\mathbf{u}_0$ can be expressed as $\alpha \mathbf{u}$, where the parameter $\alpha \in [0, 1]$ is an arbitrary value that controls the degree of Eulerian and Lagrangian nature.

By using the definition given in Equation (16), Equations (12)–(15) can be rewritten as follows in the ALE framework.

$$\frac{d\rho}{dt} + (1 - \alpha) \mathbf{u} \cdot \nabla \rho = -\rho \nabla \cdot \mathbf{u} \quad (17)$$

$$\frac{d\mathbf{u}}{dt} + (1 - \alpha) (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{\nabla P}{\rho} + \frac{\nabla \boldsymbol{\tau}^v}{\rho} + \mathbf{f}_b \quad (18)$$

$$\frac{dT}{dt} + (1 - \alpha) \mathbf{u} \cdot \nabla T = -\frac{k}{c_p} \nabla^2 T \quad (19)$$

$$P = F(\rho) \quad (20)$$

If α approaches zero, Equations (17)–(20) transforms into the fully Eulerian formalism expressed in Equations (12)–(15). By contrast, if α approaches one, Equations (17)–(20) transforms into the fully Lagrangian formalism expressed in Equations (6)–(9).

2.3 | ALE-SPH governing equations

By using the SPH kernel approximation, the governing equations based on the ALE framework can be written as follows:

$$\frac{d\rho_i}{dt} = -\rho_i \sum_j (\mathbf{u}_j - \mathbf{u}_i) \cdot \nabla_i W_{ij} \frac{m_j}{\rho_j} + D_i^\rho + C_{1,i}, \quad (21)$$

$$\frac{d\mathbf{u}_i}{dt} = -\frac{1}{\rho_i} \sum_j (p_i + p_j) \nabla_i W_{ij} \frac{m_j}{\rho_j} + \mu_i K \sum_j \Pi_{ij} \nabla_i W_{ij} \frac{m_j}{\rho_j} + \mathbf{f}_b + \mathbf{C}_{2,i}, \quad (22)$$

$$\frac{dT_i}{dt} = -\frac{k}{c_p} \nabla^2 T_i + C_{3,i}, \quad (23)$$

$$P_i = F(\rho_i), \quad (24)$$

where μ_i denotes dynamic viscosity, $K = 2(d + 2)$ and d denote spatial dimensions. The coefficients D_i^ρ , $C_{1,i}$, $\mathbf{C}_{2,i}$, $C_{3,i}$ are defined as:

$$D_i^\rho = \delta h c_0 \frac{\rho_0}{\rho_i} \sum_j \psi_{ij} \cdot \nabla_i W_{ij} \frac{m_j}{\rho_j}, \quad (25)$$

$$C_{1,i} = (\alpha_i - 1) \mathbf{u}_i \cdot \sum_j (\rho_j - \rho_i) \nabla_i W_{ij} \frac{m_j}{\rho_j}, \quad (26)$$

$$\mathbf{C}_{2,i} = (\alpha_i - 1) \mathbf{u}_i \cdot \sum_j ((\mathbf{u}_j - \mathbf{u}_i) \otimes \nabla_i W_{ij}) \frac{m_j}{\rho_j}, \quad (27)$$

$$C_{3,i} = (\alpha_i - 1) \mathbf{u}_i \cdot \sum_j (T_j - T_i) \nabla_i W_{ij} \frac{m_j}{\rho_j}. \quad (28)$$

In Equation (25), D_i^ρ is the density diffusion term proposed by Antuono et al.¹⁰ for relieving high-frequency pressure noise. The parameter δ is generally set to 0.1, regardless of resolutions and problems,¹⁰ and the term ψ_{ij} is written as follows:

$$\psi_{ij} = 2 \left\{ (\rho_j - \rho_i) - \frac{1}{2} \left(\langle \nabla \rho \rangle_j^L + \langle \nabla \rho \rangle_i^L \right) \cdot (\mathbf{r}_j - \mathbf{r}_i) \right\} \frac{\mathbf{r}_j - \mathbf{r}_i}{\|\mathbf{r}_j - \mathbf{r}_i\|^2}. \quad (29)$$

The subscript L indicates that the gradient is reformulated using renormalization matrix $\mathbf{L}_i$, and it is expressed as follows:

$$\langle \nabla \rho \rangle_i^L = \sum_j (\rho_j - \rho_i) \mathbf{L}_i \nabla_i W_{ij} V_j; \mathbf{L}_i = \left[\sum_j (\mathbf{r}_j - \mathbf{r}_i) \otimes \nabla_i W_{ij} V_j \right]^{-1}, \quad (30)$$

where $\mathbf{L}_i$ is the renormalization matrix, and the operator $\otimes$ denotes the dyadic product. Hereinafter, the density diffusion term is written as D_i^ρ . The third term on the right-hand side of the momentum equation is the viscous force term, and there are generally two choices for Π_{ij} : Morris-type viscous force Π_{ij}^{MEA} and Monaghan-type viscous force Π_{ij}^{MG} . The Monaghan-type viscous force Π_{ij}^{MG} does not dissipate near the free surface, and it can lead to a convergent solution.²¹ In this study, we consider the free surface problem, and for this reason, we adopt the Monaghan-type viscous force term:

$$\Pi_{ij} = \frac{(\mathbf{u}_j - \mathbf{u}_i) \cdot (\mathbf{r}_j - \mathbf{r}_i)}{\|\mathbf{r}_j - \mathbf{r}_i\|^2}. \quad (31)$$

Equations (21)–(23) include three additional terms to account for the ALE nature: $C_{1,i}$, $\mathbf{C}_{2,i}$, and $C_{3,i}$. If α_i approaches one, Equations (21)–(23) becomes identical to the well-known WCSPH formulation. This indicates that the particle i has a pure Lagrangian nature. By contrast, if α_i approaches zero, Equations (21)–(23) becomes identical to the fully Eulerian SPH formulation proposed by Lind and Stansby.¹⁷ This indicates that particle i has a pure Eulerian nature. This form was reported to converge readily with high orders of spatial accuracies.¹⁷ In the proposed ALE scheme, α_i can be specified differently for each particle in an arbitrary manner. In addition, it can be varied with time depending on the predefined conditions and criteria.

3 | SIMPLE EULERIAN-LAGRANGIAN SCHEME FOR WEAKLY COMPRESSIBLE SPH (EL-WCSPH)

3.1 | Potential violation issue of unity condition in the ALE framework

The ALE-SPH governing equations are well formulated, as shown in Equation (24), an intrinsic problem must be addressed before implementation (Figure 1). In the fully Lagrangian framework ($\alpha_i = 1.0$), the transport velocity is equal to the physical velocity. The mass of each particle gets invariant, while the volume of the particle ($V_{j,t} = m_{j,0}/\rho_{j,t}$) varies with density, where subscript t and 0 indicate the instant and initial property. By contrast, in the fully Eulerian framework ($\alpha_i = 0.0$), the particle is fixed, and the transport velocity is included in the advection term so, the particle volume is fixed, but its mass changes due to convection ($m_{j,t} = \rho_{j,t}V_{j,t}$). In the ALE framework ($0 < \alpha_i < 1$), the particle moves according to the part of the physical velocity and the residual part is included in the convection. Both the mass and volume of each particle can be thought to vary during the simulation ($m_j^t = \rho_j^t V_j^t$). Therefore, correction and update of the particle mass and volume should be conducted for each time-step. This issue can be neglected in the ISPH-based ALE scheme,^{17,18} which enforces a constant volume across the entire domain by solving PPEs. However, it may not be trivial for WCSPH, which allows particle compressibility. If a large compressibility is involved in the physics, such as a shock wave problem, these errors might be significant. In this respect, Oger et al.¹² calculated the change in volume at each step by solving the volume flux equation derived using the Reynolds transport theorem. Antuono et al.¹⁶ reshaped the volume flux equation and calculated the variation of mass in each step. The mass flux diffusive term was added to main stability of the scheme. In this study, we propose a simple and straightforward method of mass correction without solving additional diffusive terms, and the main idea is illustrated in Figure 2.

Figure 2 shows the issue of potential violation of the SPH kernel unity condition in the ALE framework, unless the mass and volume of the particles are not corrected during the simulation. In the ALE framework, the transport velocity can be varied and the particle positions become disordered depending on the value of α_i , despite the physical velocity being uniform in space. Therefore, although the density of the particle is well estimated using Equation (21), the mass and volume of the particle must be updated at the new particle location. Failing this, the unity condition of the SPH kernel approximation cannot be satisfied.

3.2 | Particle mass correction

The violation of the unity condition issue has been also discussed in Lagrangian framework in the presence of shocks and density discontinuities.^{22–26} In their studies, they adopted corrected volume terms instead of general volume element

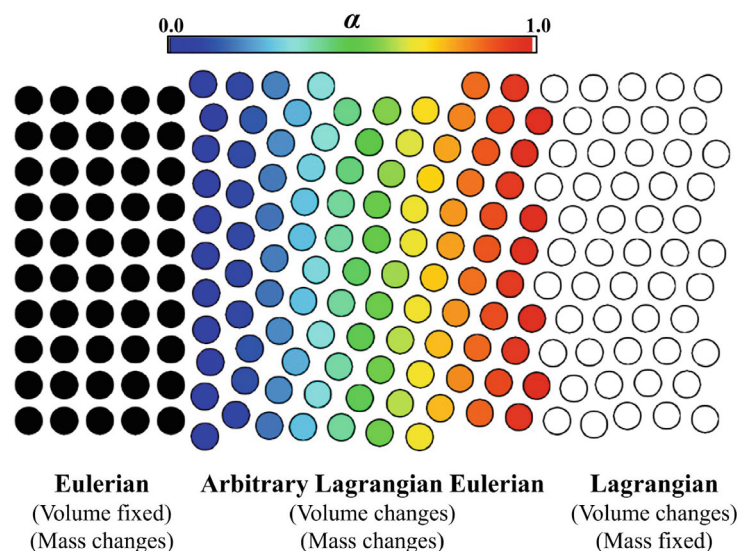


FIGURE 1 Different motion of SPH particles at uniform velocity according to the parameter α

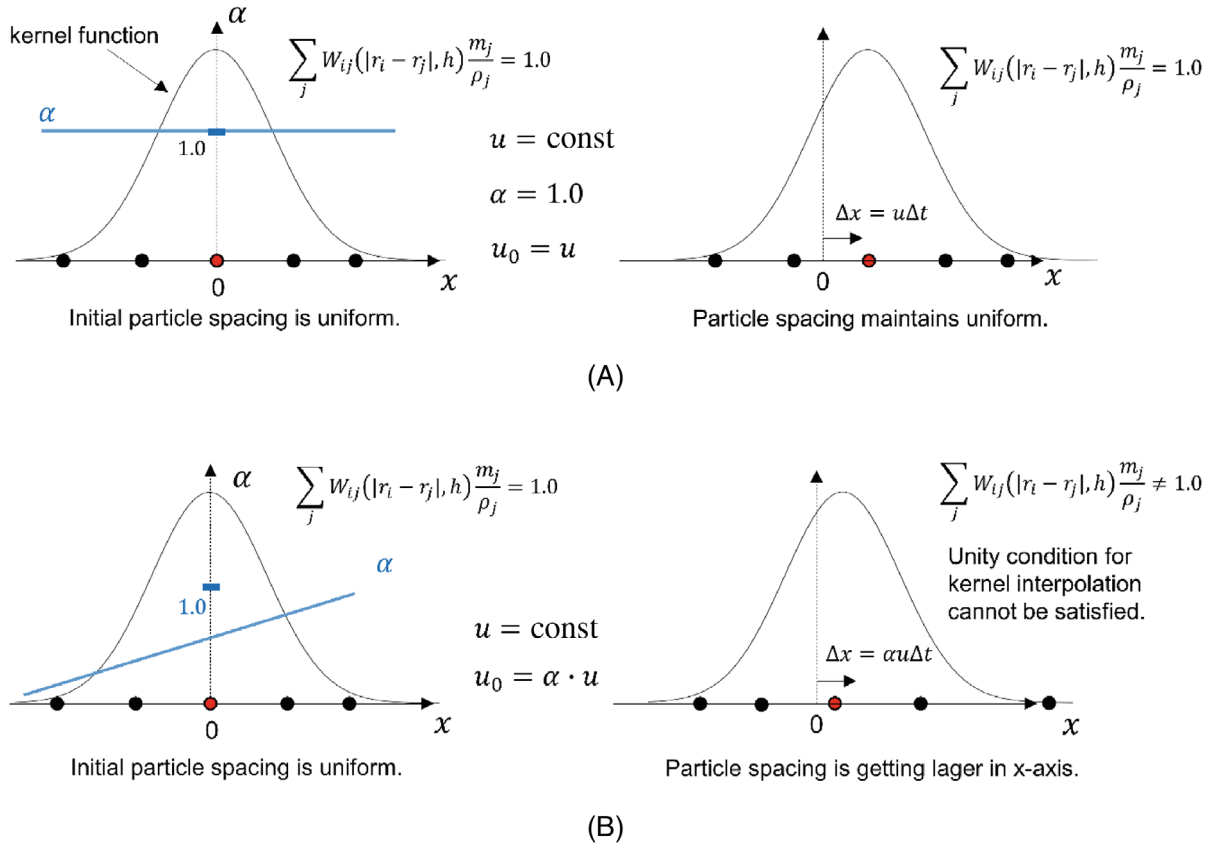


FIGURE 2 Potential violation issue of unity condition in the ALE framework without particle mass correction. (A) Particles move with the same physical and transport velocity in the fully Lagrangian framework; (B) Particles move with the same physical velocity but varying transport velocity in the ALE framework

($V_{j,t} = m_{j,0}/\rho_{j,t}$) to satisfy the unity condition. In our study, we focused on the violation of the unity condition which can arise in different transport velocity distribution of particles, and the mass is corrected before all of calculation procedure to get appropriate volume which satisfies unity condition ($V_{j,t} = m_{j,t}/\rho_{j,t}$). We simply corrected the mass of each particle ($m_{i,t}$) in each time-step by using instant density ($\rho_{i,t}$) and instant particle number density ($1/V_{i,t}$), as follows:

$$m_{i,t} = \rho_{i,t} V_{i,t} = \frac{\rho_{i,t}}{\langle \frac{1}{V_i} \rangle_t} = \frac{\rho_{i,t}}{\sum W_{ij}}. \quad (32)$$

The particle near the wall boundaries or free surface cannot satisfy the kernel unity condition ($\sum (m_j/\rho_j) W_{ij} = 1$) due to the particle deficiency. Therefore, the Shepard filter is utilized to compensate for the particle deficiency.

$$m_i = \frac{\rho_i \sum \frac{m_j}{\rho_j} W_{ij}}{\sum W_{ij}}. \quad (33)$$

However, the initial volume may not be uniform in some cases. In these cases, the mass can be obtained by the normalized form of Equation (32) based on the initial volume ($V_{0,i} = m_{0,i}/\rho_{0,i}$), where $m_{0,i}$ and $\rho_{0,i}$ indicate the initial mass and density of i th particle. Due to this modification on mass correction, the multi-resolution is implementable equally but it is not covered deeply in this study.

$$m_i = \rho_i \langle V_i \rangle = \rho_i V_{0,i} \langle V_i \rangle = \rho_i V_{0,i} \frac{\sum \frac{m_j}{\rho_j} W_{ij}}{\sum W_{ij}} \frac{\sum \frac{m_j}{\rho_j} \frac{1}{V_{0,j}} W_{ij}}{\sum \frac{m_j}{\rho_j} W_{ij}} = \rho_i V_{0,i} \frac{\sum \frac{m_j}{\rho_j} \frac{1}{V_{0,j}} W_{ij}}{\sum W_{ij}}. \quad (34)$$

Therefore, governing equations of the proposed EL-WCSPH scheme are constituted as follows:

$$\frac{d\rho_i}{dt} = -\rho_i \sum_j (\mathbf{u}_j - \mathbf{u}_i) \cdot \nabla_i W_{ij} \frac{m_j}{\rho_j} + D_i^\rho + C_{1,i}, \quad (35)$$

$$\frac{d\mathbf{u}_i}{dt} = -\frac{1}{\rho_i} \sum_j (p_i + p_j) \nabla_i W_{ij} \frac{m_j}{\rho_j} + \mu_i K \sum_j \Pi_{ij} \nabla_i W_{ij} \frac{m_j}{\rho_j} + \mathbf{f}_b + \mathbf{C}_{2,i}, \quad (36)$$

$$\frac{dT_i}{dt} = -\frac{k}{c_p} \nabla^2 T_i + C_{3,i}, \quad (37)$$

$$m_i = \frac{\rho_i V_{0,i} \left(\sum_j \frac{m_j}{\rho_j} \frac{1}{V_{0,j}} W_{ij} \right)}{\left(\sum_j W_{ij} \right)}, \quad (38)$$

$$P_i = F(\rho_i). \quad (39)$$

3.3 | Particle disorder correction

In the mixed Eulerian–Lagrangian scheme, the Lagrangian moving particles can penetrate through the fixed Eulerian region and create significant particle disorder if proper numerical treatment is not applied. In addition, the particle disorder is main cause of the error in SPH approximation. In this study, a PST^{26,27} is adopted to alleviate the irregular particle distribution and prevent the penetration of Lagrangian particles into the Eulerian region. The positions of the particles are relocated according to their distribution based on Fick's law and the von Neumann stability analysis:

$$\delta \mathbf{x}_i = -D_i \nabla_i C_i = -D_i \sum_j \left[1 + 0.2 \left(\frac{W_{ij}}{W_{ij}(\Delta x)} \right)^4 \right] \frac{m_j}{\rho_j} \nabla_i W_{ij},$$

$$D_i = 0.5h_i^2, \quad (40)$$

where D_i is the shifting coefficient, and C_i is the particle concentration. In this study, the shifting coefficient provided by Fournakos et al.¹⁸ is used. As the particles are realigned, their physical properties such as velocity and density are interpolated using the following Taylor series:

$$\varphi_{i'} = \varphi_i + (\nabla \varphi)_i \cdot \delta \mathbf{r}_{ii'} + O(\delta \mathbf{r}_{ii'}^2), \quad (41)$$

where φ is the physical property of particle i , and $\delta \mathbf{r}_{ii'}$ is the shifting vector between the original position ($\mathbf{r}_i$) and the relocated position ($\mathbf{r}_{i'}$).

Controlling particle shifting in the vicinity of the free surface is problematic. Nonphysical particle shifting due to incorrect calculation of the concentration gradient near the free surface can affect global calculation accuracy. However, shifting only the inner particles leads to numerical inconsistency and particle clustering near the free surface. Therefore, several methods are proposed to apply Fick's law to free surface particles. Lind et al.²⁷ allowed slight shifting in the vertical direction to the free surface and controlled with a parameter according to the problem. Khayyer et al.²⁸ and Sun et al.²⁹ calculated the normal vector at the free surface and completely prevented particles to shift across the vertical direction of the free surface. The difference between two research is how to define free surface particles and calculate normal vector. Khayyer et al.²⁸ define the free surface particles with the criterion of the divergence of position vector and calculate normal vector based on gradient of concentration ($\nabla_i C_i$). Sun et al.²⁹ determined the free surface particles based on the criterion of the minimum eigenvalue of renormalization matrix $\mathbf{L}_i$. The normal vector is calculated based on the gradient of minimum eigenvalue. In this study, the optimized particle shifting (OPS) technique by Khayyer et al.,²⁸ which is one of the famous shifting method on free surface is adopted. First, the free surface and the neighboring particles are detected using the criterion of the divergence of the position vector ($\nabla \cdot \mathbf{r}$), as in Figure 2, which is similar to Lind et al.²⁷ and Khayyer et al.²⁸ (Figure 3).

After particle detection, the corrected normal vectors of the free surface and vicinity particles are calculated as follows:

$$\tilde{\mathbf{n}}_i = -\frac{\mathbf{B}_i \cdot \nabla C_i}{|\mathbf{B}_i \cdot \nabla C_i|} \quad (42)$$

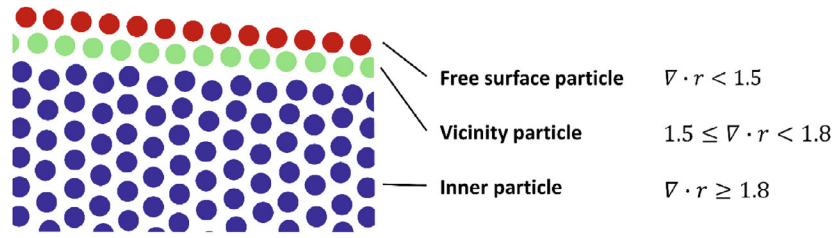


FIGURE 3 Different types of particles according to divergence criterion in OPS algorithm

The particle shifting displacement vector near the free surface can be expressed as follows, where the normal component of particle shifting distance is completely removed:

$$\delta \mathbf{r}_i = -D_{\text{shift}} \nabla_S C_i = -D_{\text{shift}} (\nabla C_i - \nabla_N C_i) = -D_{\text{shift}} (\mathbf{I} - \tilde{\mathbf{n}}_i \otimes \tilde{\mathbf{n}}_i) \nabla C_i \quad (43)$$

where $\nabla_S C_i$ and $\nabla_N C_i$ denote the tangential and normal components of the gradient of the concentration, respectively. The physical variables (density, temperature, and velocity) are interpolated in a manner similar to the inner particles.²⁸

3.4 | Assessment of proposed EL-WCSPH scheme

To validate the EL-WCSPH scheme, the isentropic 1D shock tube problem which is brought from Marongiu.³⁰ Similar to Collé et al.,¹⁴ a stable solution was obtained in Eulerian framework with the presence of artificial viscosity and diffusion term. The governing equations for simple shock problem are as follows.

$$\frac{d\rho_i}{dt} = -\rho_i \sum_j (\mathbf{u}_j - \mathbf{u}_i) \cdot \nabla_i W_{ij} \frac{m_j}{\rho_j} + D_i^p + C_{1,i} \quad (44)$$

$$\frac{d\mathbf{u}_i}{dt} = -\frac{1}{\rho_i} \sum_j (p_i + p_j) \nabla_i W_{ij} \frac{m_j}{\rho_j} + \alpha_{av} h c_0 \sum_j \Pi_{ij} \nabla_i W_{ij} \frac{m_j}{\rho_j} + \mathbf{f}_b + \mathbf{C}_{2,i} \quad (45)$$

$$m_i = \rho_i V_{0,i} \left(\sum_j \frac{m_j}{\rho_j} \frac{1}{V_{0,j}} W_{ij} \right) / \left(\sum_j W_{ij} \right) \quad (46)$$

$$P_i = F(\rho_i) \quad (47)$$

where, α_{av} and δ are set as 1.5 and 0.05. In order to validate mass correction, the shock problem is solved with and without mass correction. If the mass variation is considered, it is called MV scheme and the governing equations are similar to Lind et al.¹⁷ If the mass is constant, it is called MC scheme. For example, if the Equation (46) is not included and solved in Eulerian framework (transport velocity is zero for whole region), it will be called ESPH-MC. We chose the case involves a pressure discontinuity generated by the following discontinuous density field.

$$\begin{cases} \rho_L = 1100 \\ v_L = 0 \end{cases} \quad \text{and} \quad \begin{cases} \rho_R = 1000 \\ v_R = 0 \end{cases} \quad (48)$$

L and R indicate the left and right region of the domain. The reference density and speed of sound are chosen to be $\rho_0 = 1000 \text{ kg/m}^3$ and $c_0 = 1466 \text{ m/s}$. The shock tube is taken with the length of 0.1 m and the discontinuity is initially defined at center. The initial particle space is set as $5 \mu\text{m}$. The Figure 1 shows the pressure/velocity distribution at 10^{-5} s using ESPH-MV scheme (Figure 4).

In this case, the maximum compressibility of density during the measurement time is around 10%. It is difficult to solve with weakly compressible assumption and the mass cannot be assumed to be keep constant. The Figure 5 shows the comparison of simulation results of ESPH-MC and ESPH-MV schemes for the same problem. Without the mass correction, it is impossible to accurately predict the pressure and velocity distribution of shocks generated by discontinuities. The same problem is solved with different transport velocity by three cases. In the first and second case, the transport

velocity is set as $1/10(\alpha = 0.1)$ and $1/2(\alpha = 0.5)$ of the physical velocity. In the third case, the interface (center $x = 0$) is set as complete Lagrangian ($\alpha = 0$), both ends ($x = \pm 0.05$) are set as Eulerian ($\alpha = 1$), and the transport velocity varies linearly ($\alpha = 0-1$) between the end sides and interface (Figure 6).

When the particles move, the non-uniform distribution is induced and such distribution induces numerical noise at the interface. Therefore, the particle disorder correction is applied when the transport velocity is larger than 0. In the Figure 7, the results for 4 schemes with or without mass correction and with or without particle disorder correction are compared. The results with particle disorder correction is remarked by attaching PST after the name of scheme.

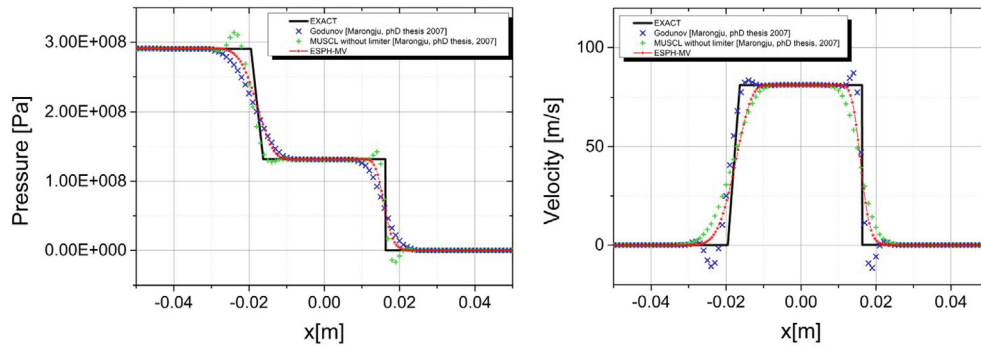


FIGURE 4 Comparison between the pressure and velocity fields of ESPH-MV with other Riemann based SPH results and exact solution.

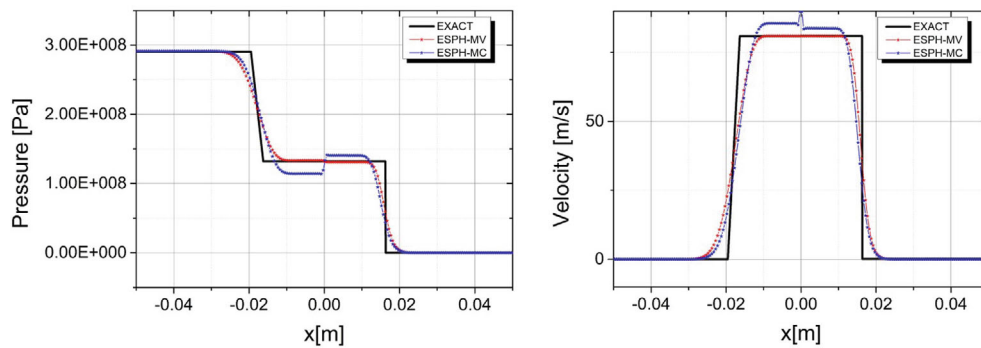


FIGURE 5 Comparison between the pressure and velocity fields of ESPH-MV and ESPH-MC schemes

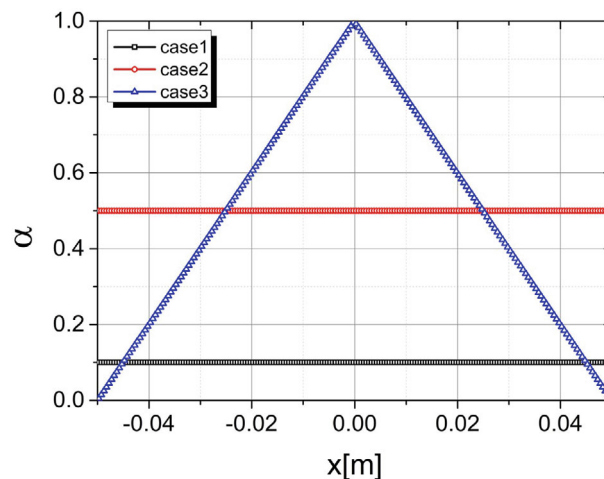


FIGURE 6 Three different simulation conditions of transport velocity parameter α

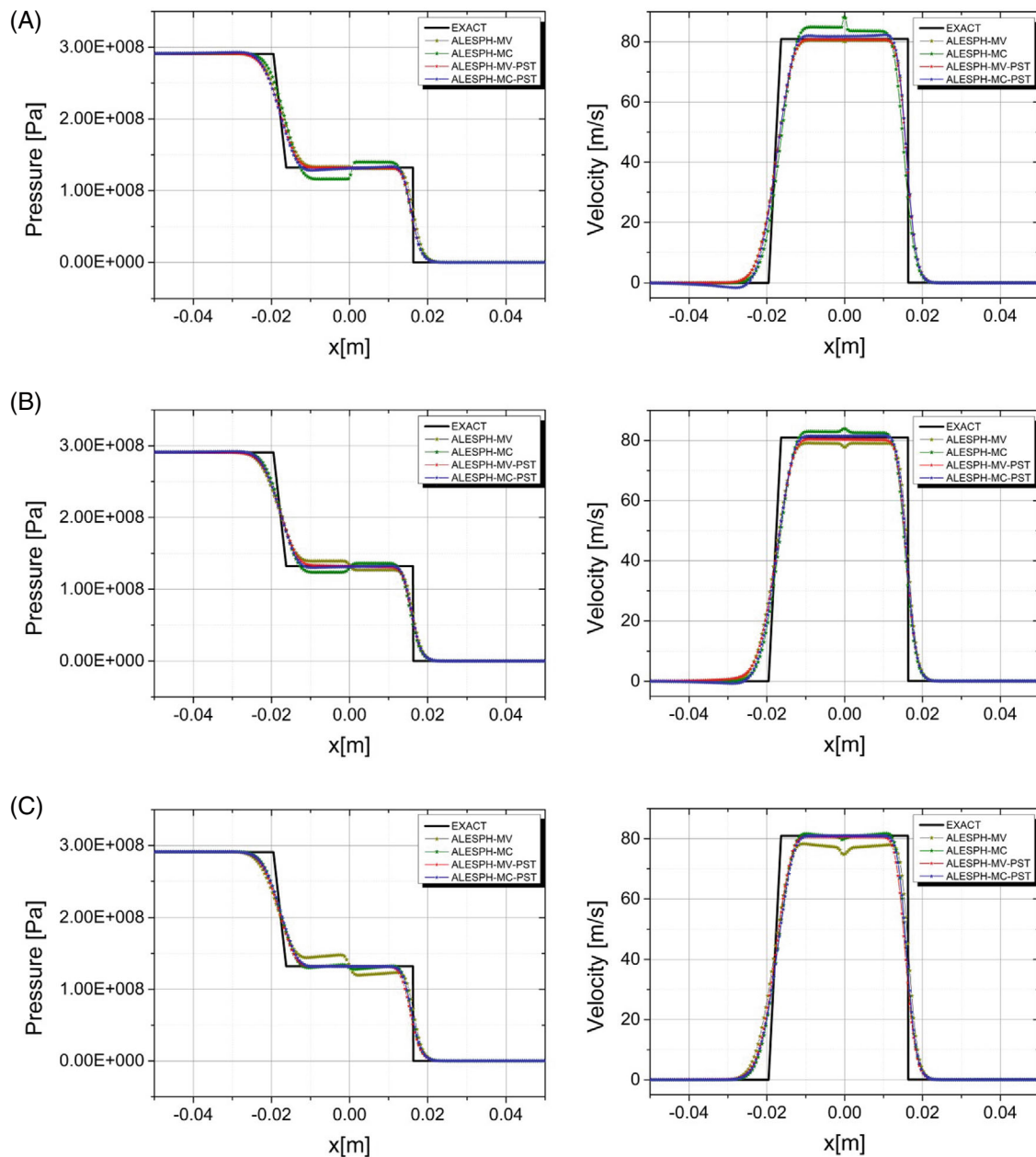


FIGURE 7 Comparison of the pressure and velocity fields between ALESPH schemes with three different transport velocity parameter α (A) Case 1 (B) Case 2 (C) Case 3

In the Figure 8, the results with particle disorder correction are shown to represent the pressure and velocity distribution in good agreement to exact solution. Next, we checked whether each scheme satisfies the unity condition in three cases. The unity condition is shown to be most satisfied when the particle disorder correction and particle mass correction is both used for all cases.

4 | IMPLEMENTATION

4.1 | Time marching

In this study, the governing equations, particle mass correction, and particle disorder correction in Sections 2.1–2.3 are implemented as shown in Figure 9. The overall calculation procedure of the Eulerian–Lagrangian SPH (ELSPH) is almost

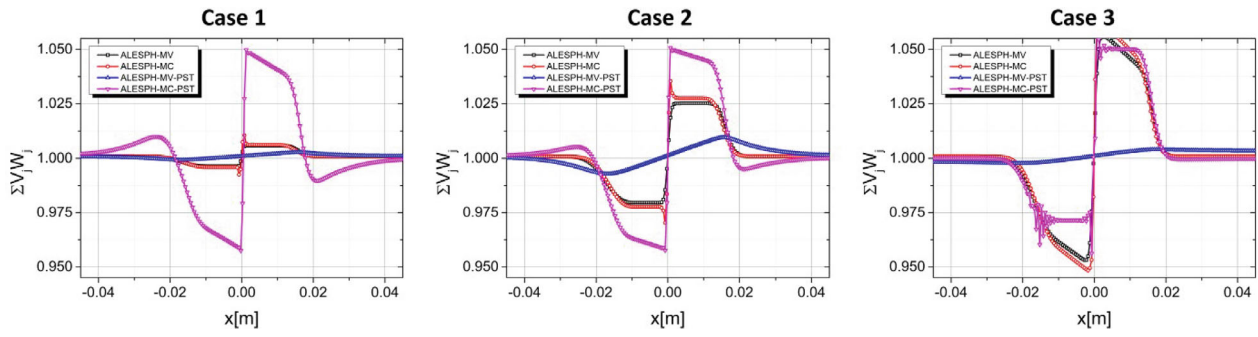


FIGURE 8 Assessment of the unity condition for ALESPH schemes with three different transport velocity parameter α

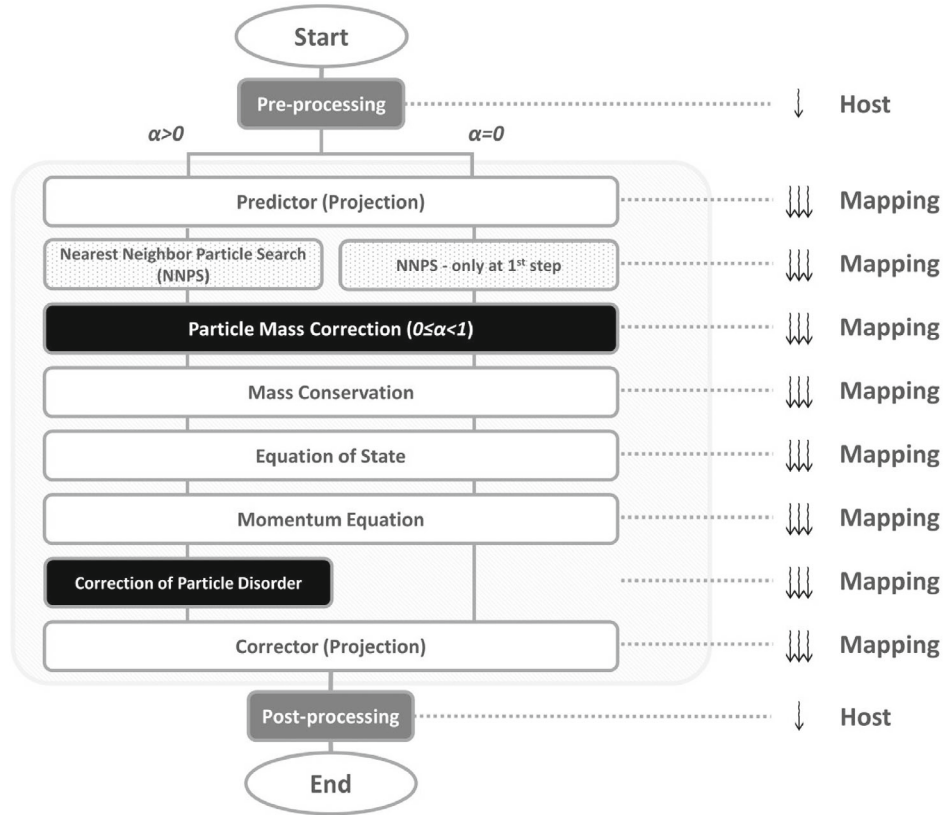


FIGURE 9 Algorithm of mixed Eulerian-Lagrangian SPH (EL-SPH).

identical to the standard WCSPH. However, the mass and particle disorder correction steps are included additionally. As shown in Figure 9, mass correction is performed after the predictor step in time integration, and particle disorder correction is performed after the corrector step in time integration and update. In this study, the following predictor-corrector time integration scheme is used.

$$\mathbf{u}_{t+\frac{\Delta t}{2}}^p = \mathbf{u}_t + \frac{\Delta t}{2} \left(\frac{d\mathbf{u}}{dt} \right)_t \quad (49)$$

$$\mathbf{r}_{t+\frac{\Delta t}{2}}^p = \mathbf{r}_t + \alpha \frac{\Delta t}{2} \left(\frac{d\mathbf{r}}{dt} \right)_t \quad (50)$$

$$\rho_{t+\frac{\Delta t}{2}}^p = \rho_t + \frac{\Delta t}{2} \left(\frac{d\rho}{dt} \right)_t \quad (51)$$

where t and Δt denote time and time step, respectively. The superscript p denotes “predictor.” The time derivatives of position, velocity, density, and energy are reevaluated by solving the discretized SPH governing equations based on the predicted values. Then, the field variables are reintegrated over the entire time step by using the updated time derivatives in the correction step.

$$\mathbf{u}_{t+\Delta t}^c = \mathbf{u}_t + \Delta t \left(\frac{d\mathbf{u}}{dt} \right)_{t+\frac{\Delta t}{2}} \quad (52)$$

$$\mathbf{r}_{t+\Delta t}^c = \mathbf{r}_t + \alpha \Delta t \left(\frac{d\mathbf{r}}{dt} \right)_{t+\frac{\Delta t}{2}} \quad (53)$$

$$\rho_{t+\Delta t}^c = \rho_t + \Delta t \left(\frac{d\rho}{dt} \right)_{t+\frac{\Delta t}{2}} \quad (54)$$

where the superscript c denotes “corrector.” These corrected values serve as the initial values in the next timestep. In the ALE framework, the particle positions are updated based on the transport velocity, and therefore, the RHSs of Equations (23) and (26) are multiplied with the parameter α . Since the SPH is generally based on explicit time integration schemes, the maximum timestep is limited by several factors to ensure numerical stability.

$$\Delta t_i \leq \min \left(0.2 \min \frac{h_i}{c_0}, 0.125 \frac{h_i^2}{\nu_i} \right) \quad (55)$$

where, the ν is the kinematic viscosity. The speed of sound c_0 is set to be $c_0 \geq 10U_{\max}$ to enforce the compressibility lower than 1%, where $U_{\max}$ is anticipated maximum velocity.

4.2 | GPU implementation

In this study, the fluid flow and heat transfer are solved in the Eulerian, Lagrangian, and Eulerian–Lagrangian framework using a single GPU-based algorithm presented in Figure 9. A GPU is a specialized processor that was originally designed to accelerate the graphics of three-dimensional (3D) rendering. A GPU consists of a number of streaming multiprocessors (SM), which themselves are composed of dozens of CUDA cores. Therefore, GPUs can process multiple pieces of data simultaneously. The preprocessing step (read input file, initializing parameters, etc.) and post-processing (save output) step are implemented by CPU (host), while the entire calculation processes are parallelized by the GPU (device). All the simulations in this study have been run on a workstation equipped with NVIDIA Tesla V100 PCIE 32GB and Intel® Xeon® Silver 4210R CPU@2.40GHz.

In the entire calculation steps, the nearest neighbor particle search (NNPS) is the most time-consuming process. In this study, a cell-linked list NNPS algorithm is implemented in the GPU (for the detailed information on GPU implementation, we recommend readers to read References 5 and 31). In the case where the transport velocity is zero and particles are fixed (Eulerian), the NNPS is performed only at first step. However, if the transport velocity is larger than zero and particles move, the neighbor particles are altered every step and the NNPS is required for every step. For the SPH interpolation and time integration steps, a parallel mapping algorithm is used with parallel reduction. The particle disorder correction step is activated when the transport velocity is larger than zero. As a result, the entire calculation processes are parallelized by the GPU, except for file input/output (IO).

5 | BENCHMARK SIMULATIONS

In this section, various benchmark simulations are conducted to validate the proposed ALE formulations and framework in terms of accuracy and robustness. The benchmark cases include several steady and transient 2D hydrodynamics problems: (1) pipe flow, (2) lid-driven flow, (3) Taylor–Green vortices, (4) vortex shedding, (5) sloshing. Furthermore, complicated hydrodynamic flows of (6) vortex shedding on a free surface and (7) natural convection in a square cavity which includes convective heat transfer are simulated. For the all benchmark problems, the particles are initially arranged

in a uniform distribution. The initial configuration and number of particles for all benchmarks are categorized in the Figure 10 and Table 1. In the following sections, we summarize and discuss our results.

5.1 | 2D pipe flow

The benchmark simulation is conducted for a two-dimensional (2D) pipe flow problem (Figures 11 and 12). The dummy particles are allocated on the wall boundary with the no-slip condition.⁵¹ The Neumann boundary condition ($\partial p / \partial n = 0$) is enforced at the wall boundary for pressure, as proposed by Adami et al.⁵¹ In addition, open boundary conditions are imposed at the inlet and outlet, as proposed by Tafuni et al.⁵² The Reynolds number (Re), defined as $Re = \rho U D / \mu$, is 80, where U is the average flow velocity, and D is the pipe width. The speed of sound is set as $c_0 = 50 U_{\max}$, where $U_{\max} = 1.5 U$ here. Simulations are conducted for three different schemes: Eulerian SPH (ESPH), Lagrangian (LSPH), mixed Eulerian–Lagrangian SPH (ELSPH). In the ESPH scheme, the parameter α is assigned as 0.0 over the entire domain, while in the LSPH scheme, it is set to 1.0. In ELSPH scheme, the particles apart within smoothing length from the wall are treated as Eulerian ($\alpha = 0$) and left parts are set as Lagrangian ($\alpha = 1$), as in Figure 11. Moreover, the results are compared with the analytical solution of the fully developed 2D pipe flow. Figure 12 compare the velocity contour at the developed region and pressure distribution along the centerline which are obtained using different schemes. As shown in the figures, the velocity trends and pressure variation at the developed region agree well with

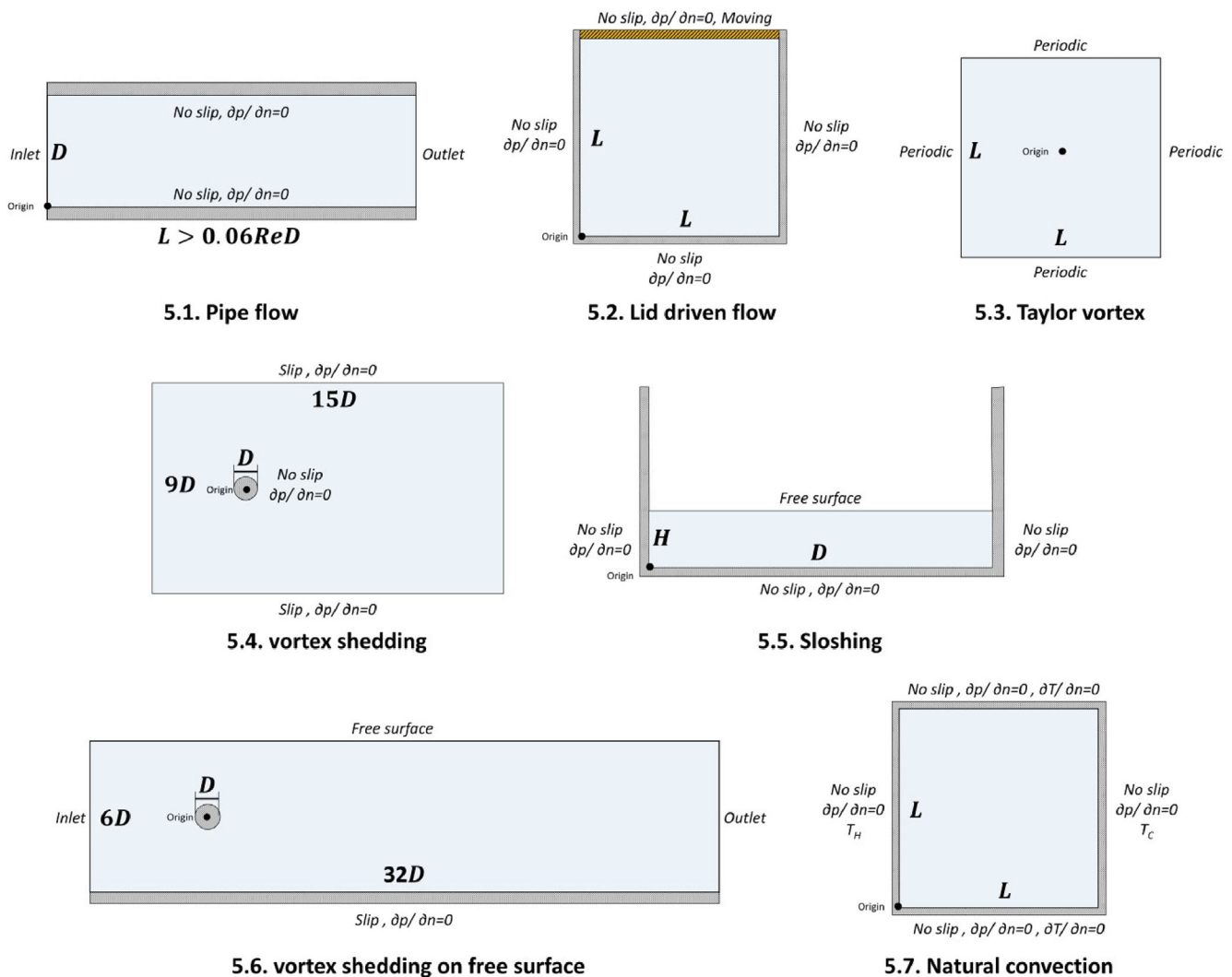


FIGURE 10 Geometric information and boundary condition in 7 benchmark

TABLE 1 Brief information of benchmark simulations

	Benchmark	Resolution	Characteristics	References
5.1	Pipe flow	$D/\Delta x = 80$	Open/steady	32
5.2	Lid-driven flow	Coarse ($L/\Delta x = 50$) Middle ($L/\Delta x = 100$) Fine ($L/\Delta x = 200$)	Closed/steady	33,34
5.3	Taylor–Green vortices	Coarse ($L/\Delta x = 50$) Middle ($L/\Delta x = 200$) Fine ($L/\Delta x = 800$)	Periodic/steady	35
5.4	Vortex shedding	Coarse ($D/\Delta x = 20$) Middle ($D/\Delta x = 40$) Fine ($D/\Delta x = 80$)	Open/transient	29,36–46
5.5	Sloshing	Coarse ($H/\Delta x = 50$) Fine ($H/\Delta x = 100$)	Free-surface/transient	47
5.6	Vortex shedding on a free surface	Coarse ($H/\Delta x = 20$) Fine ($H/\Delta x = 40$)	Free-surface/transient	48
5.7	Natural convection in a square cavity	$L/\Delta x = 400$	Closed/steady/heat transfer	49,50

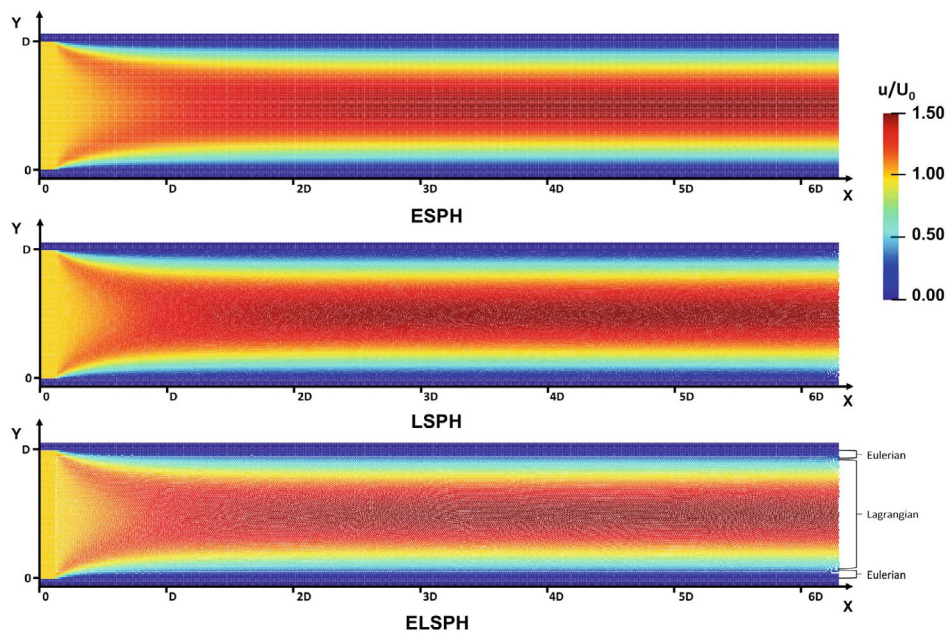


FIGURE 11 Pipe flow: Velocity magnitude contours obtained using different schemes.

the analytical solutions, regardless of the numerical schemes. The velocity distributions near the center($y = 0.5$) and near the lower wall($y = 0$) are highlighted in the figure. The maximum speed of ESPH and ELSPH are also shown to be predicted closer to the analytic solution compared to LSPH. This is due to the accuracy of the velocity distribution near the wall. ESPH and ELSPH schemes provided good predictions of velocity variation near the wall because of the uniform particle structure near the wall. LSPH also followed the trend but some deviations exist compared to other schemes.

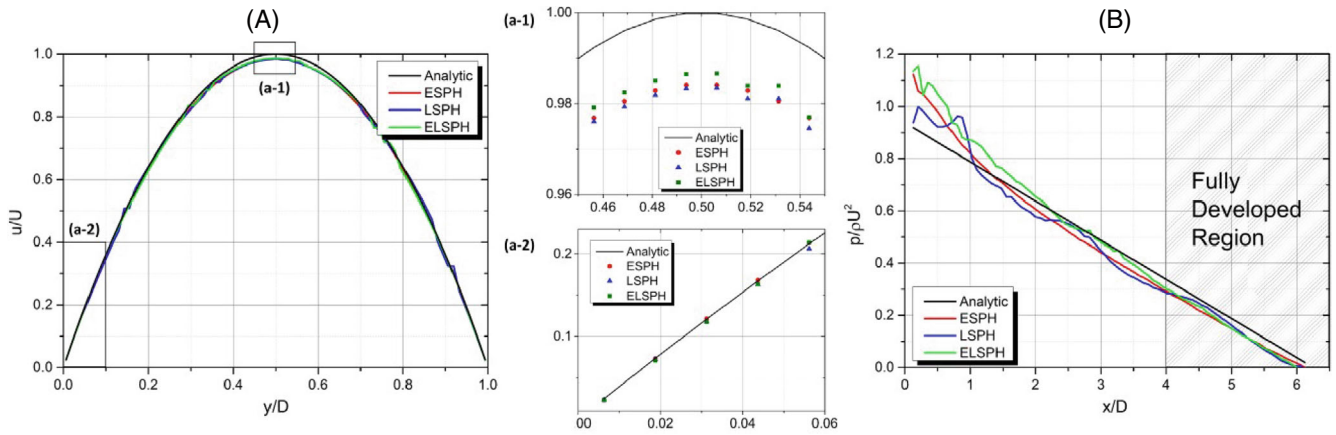


FIGURE 12 Pipe flow: (A) Velocity distribution at $x = 0.075\text{Re}D$ and (B) pressure distribution along the center line $y = 0.5$

5.2 | 2D lid-driven flow in a square cavity

The lid-driven cavity problem at $\text{Re} = 1000$ is solved as a benchmark problem in this study. In the lid-driven cavity problem, a flow is driven in a fully confined square cavity with dimensions of $[0, L] \times [0, L]$, by means of the tangential motion of a bounding wall.³³ In this simulation, the no-slip boundary condition is enforced on all side walls, and the upper wall has a constant velocity U . The simulations are conducted using the ESPH, LSPH, and ELSPH schemes with three different resolutions ($L/dx = 50, 100, 200$). The speed of sound is set as $c_0 = 10U_{\max}$, where $U_{\max}$ is the anticipated maximum velocity and equal to the velocity of lid in this problem.

In the ELSPH scheme, the cavity is divided into two regions; Eulerian and Lagrangian. The Lagrangian region ($\alpha = 1.0$) is located at the center of the cavity, as indicated by the dotted line in Figure 13, and surrounded by the Eulerian region.

In Figure 13, the dimensionless velocity contours with streamlines are compared for the three simulation cases. According to the simulations, a large vortex is driven in the center, and secondary vortices are generated at the corners. The figure shows that in all three cases the main and secondary vortices are clearly visible, regardless of the scheme.

In Figure 14, a comparison of the horizontal velocity across the vertical centerline ($x = 0.5L$) and the vertical velocity across the horizontal centerline ($y = 0.5L$) at the steady state are presented and compared with converged finite difference (FD) results³³ and Riemann-based SPH-ALE method by Oger et al.¹² The peak velocity and global trend are shown to be close and L2-norms of the overall calculation are plotted against the particle spacing dx . The L2-norm is calculated as

$$\|e\|_2 = \frac{(\sum_N |u_{\text{SPH}} - u_{\text{FD}}|^2)^{1/2}}{N} \quad (56)$$

All schemes showed converged results, but the Lagrangian case showed better results of local velocity trend for the middle ($L/dx = 100$) and high ($L/dx = 200$) resolutions. In addition, the pressure distribution along the square diagonal ($x = y$) are shown and compared with finite volume (FV) results.³⁴

The Figure 15 shows the vorticity fields evaluated with three schemes with spectral method (SM) results. The global trends of three schemes showed convergent results except the regions near the wall. The discrepancies might be induced by the lack of accuracy in gradient operator and showed convergence as the spatial resolution (L/dx) increases.

The Table 2 shows the position of the secondary vortices near the left and right corner boundary and the evaluated vorticity at the secondary vortices. The secondary vortex in the right side of boundary is indicated as BR and the vortex in the left side is written as BL. The results are compared with FD results.³³ The results of the three schemes converged to the FD results. The position of the vortices are similar but, ESPH and ELSPH predicted the vorticity more accurately than LSPH.

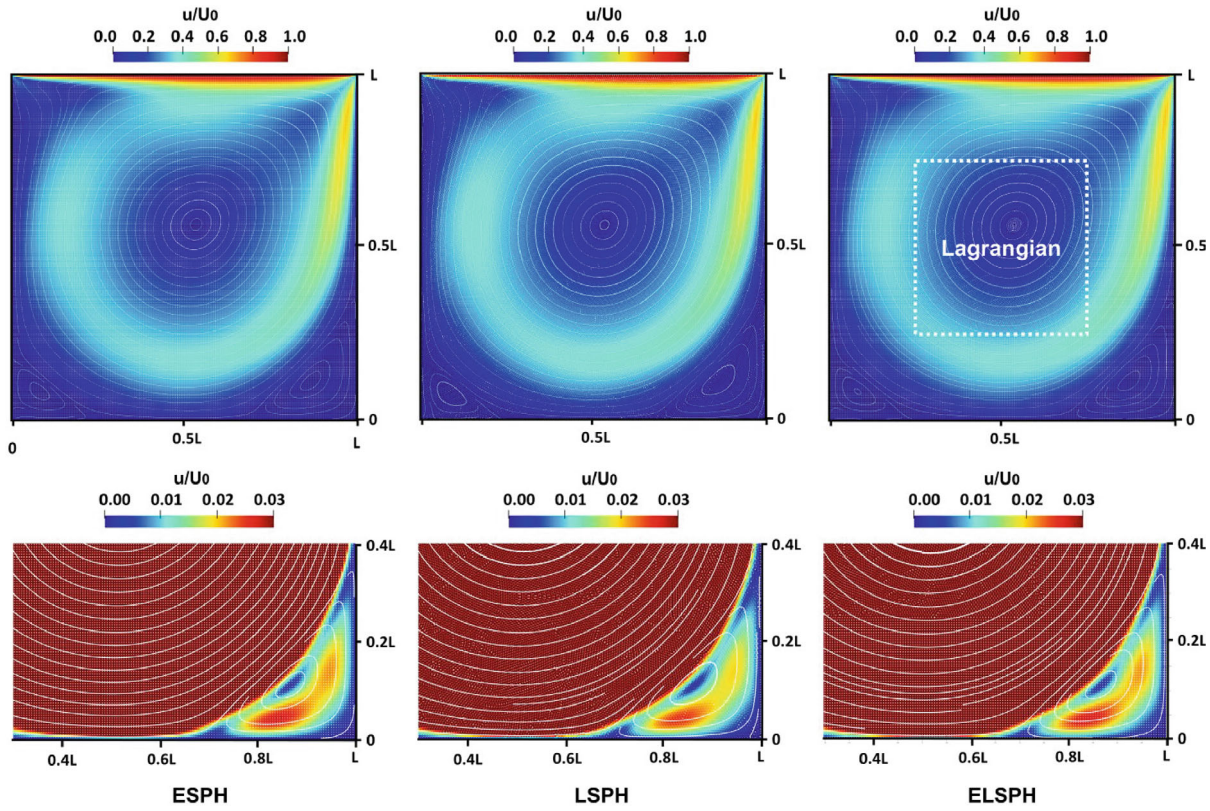


FIGURE 13 Lid driven problem: Velocity contours with streamlines (upper row) and secondary vortices (bottom row).

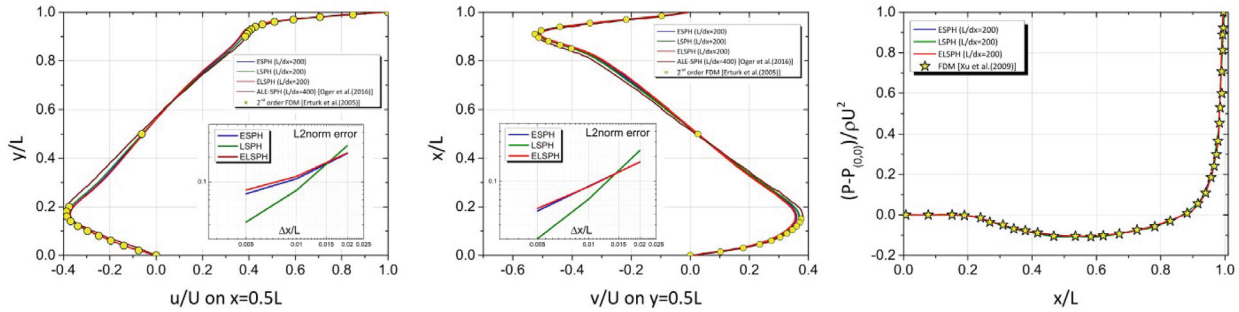


FIGURE 14 Lid-driven problem: Horizontal velocity along vertical centerline ($x = 0.5L$), vertical velocity along horizontal centerline ($y = 0.5L$), and pressure profile along square diagonal ($x = y$)

5.3 | 2D Taylor vortex

A benchmark simulation was conducted for a two-dimensional Taylor–Green vortex.³⁵ The fluid domain is unbounded with dimensions of $[-0.5L, 0.5L] \times [-0.5L, 0.5L]$, and performed at $Re = 200$ in three different spatial resolutions. The domain was initialized with four periodic counter-rotating vortices decaying by means of viscosity dissipation. The analytical solutions of the velocity and pressure fields can be expressed as follows:

$$\begin{aligned}
 u^*(x^*, y^*, 0) &= \sin(2\pi x^*) \cos(2\pi y^*) e^{-\frac{8\pi^2 t}{Re}} \\
 v^*(x^*, y^*, 0) &= -\cos(2\pi x^*) \sin(2\pi y^*) e^{-\frac{8\pi^2 t}{Re}} \\
 p^*(x^*, y^*, 0) &= (\cos(4\pi x^*) + \cos(4\pi y^*)) e^{-\frac{16\pi^2 t}{Re}}
 \end{aligned} \tag{57}$$

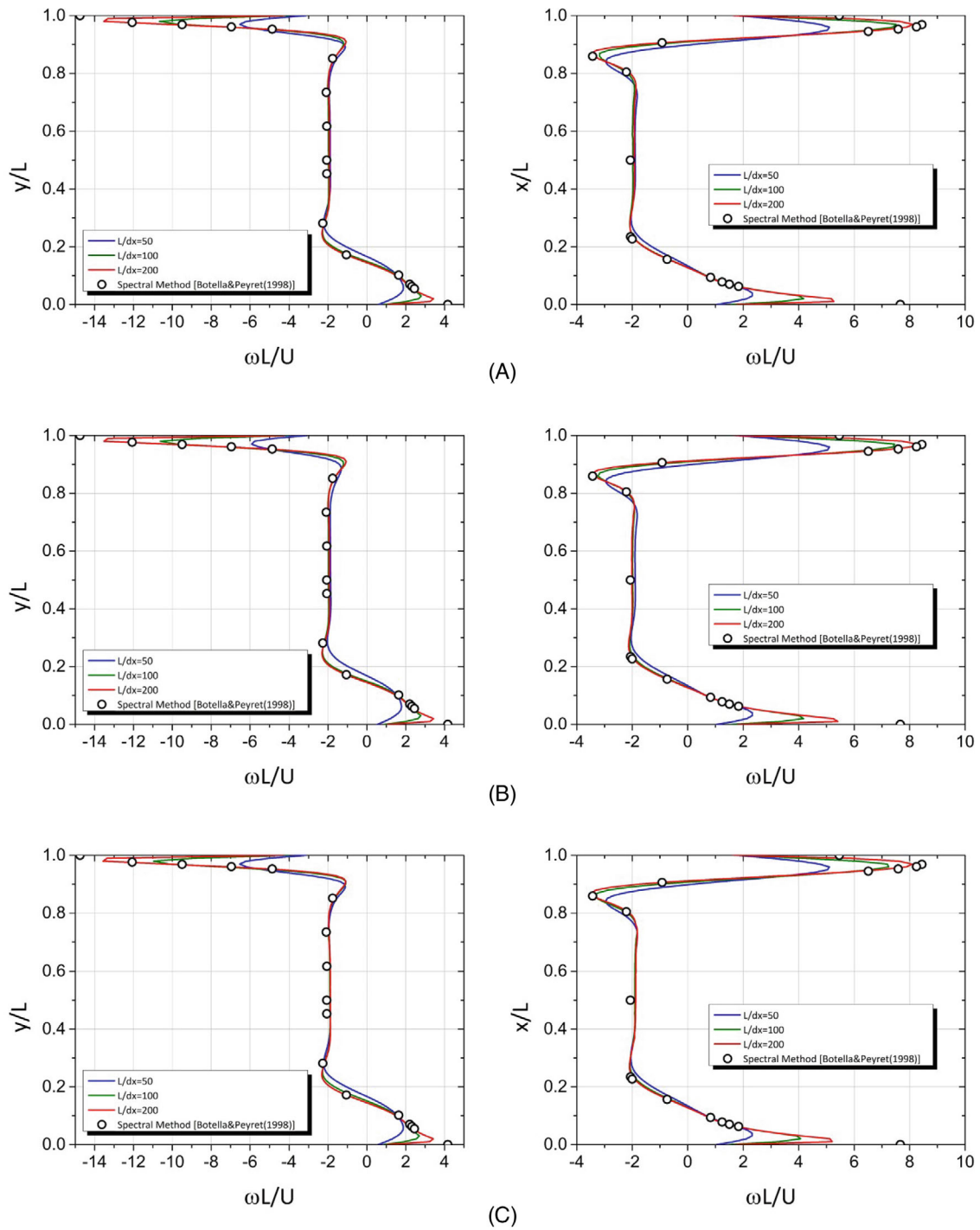


FIGURE 15 Lid driven problem: Vorticity distribution along vertical centerline ($x = 0.5L$) and vertical horizontal centerline ($y = 0.5L$). (A) ESPH; (B) LSPH; (C) ELSPH

In this study, $u^*(=u/U)$, $v^*(=v/U)$, and $p^*(=p/\rho U^2)$ are dimensionless velocity components and pressure, and $x^*(=x/L)$ and $y^*(=y/L)$ are dimensionless positions. U and L denote the characteristic velocity and length, and t denotes physical time. The vertical and horizontal velocity profiles are assigned using the Neumann condition ($u \cdot n = 0$) and the Dirichlet condition, respectively. The pressure boundary condition is enforced using the homogeneous Neumann boundary condition ($\partial p / \partial n = 0$) in all directions. This problem has complex boundary conditions to deal with the Lagrangian scheme due to buffer particles coming in and going out from all sides. For simplicity, the buffer particles are treated as Eulerian particles. Similar to the lid-driven problem, simulations are conducted using the ESPH, LSPH, and ELSPH

TABLE 2 Lid driven problem: The position and vorticity predicted by three schemes

		$L/\Delta x = 50$		
-	52	ESPH	LSPH	ELSPH
x_{BL}/L	0.0859	0.03	0.161444	0.03
y_{BL}/L	0.0781	0.11	0.025979	0.11
$\omega_{BL}L/U$	0.3618	0.16433	0.169228	0.162459
x_{BR}/L	0.8594	0.85	0.821615	0.85
y_{BR}/L	0.1094	0.11	0.101533	0.11
$\omega_{BR}L/U$	1.1549	1.01972	1.06411	1.01697
		$L/\Delta x = 100$		
-	52	ESPH	LSPH	ELSPH
x_{BL}/L	0.0859	0.075	0.090696	0.075
y_{BL}/L	0.0781	0.085	0.073683	0.085
$\omega_{BL}L/U$	0.3618	0.338664	0.339377	0.336305
x_{BR}/L	0.8594	0.865	0.864732	0.865
y_{BR}/L	0.1094	0.115	0.111932	0.115
$\omega_{BR}L/U$	1.1549	1.08716	1.02051	1.08637
		$L/\Delta x = 200$		
-	52	ESPH	LSPH	ELSPH
x_{BL}/L	0.0859	0.0775	0.082954	0.0775
y_{BL}/L	0.0781	0.0825	0.075228	0.0825
$\omega_{BL}L/U$	0.3618	0.359716	0.32068	0.356342
x_{BR}/L	0.8594	0.8675	0.862308	0.8675
y_{BR}/L	0.1094	0.1125	0.109057	0.1125
$\omega_{BR}L/U$	1.1549	1.13365	1.07355	1.13577

schemes. The speed of sound is set as $c_0 = 10U_{\max}$, where $U_{\max} = U$ here. Figure 16 depicts the non-dimensional velocity magnitude and pressure contours obtained using the three different frameworks at a resolution of $L/\Delta x = 800$ at $t^* = 1$. Close agreement exists between the three different frameworks, and the pressure fields are free of noise in the transition region. The pressure and velocity distribution along the horizontal centerline ($y^* = 0$) are compared with analytical solution for three methods (ESPH, LSPH, and ELSPH) with three different spatial resolutions ($L/\Delta x = 50, 200, 800$) in Figure 17. In all three schemes with three different resolutions, the trend of velocity are similar to analytical solution. The trend of the pressure is also similar but those are quite deviated from the analytical solution at the coarse resolution ($L/\Delta x = 50$). In order to evaluate global trend, the time evolution of L1 error of velocity and pressure are plotted in Figure 18. The L1-norm is calculated as

$$||e||_1 = \frac{\sum_N |u_{SPH} - u_{analytic}|}{N} \quad (58)$$

In the coarse resolution, the ESPH rather showed relatively large error, but the error became much smaller at the middle and fine resolution. ELSPH represent more accurate results with less error than LSPH for all resolutions. In the case of velocity, LSPH showed relatively accurate results than ESPH and ELSPH. However, the magnitudes of velocity errors are about 1/10 of the pressure error.

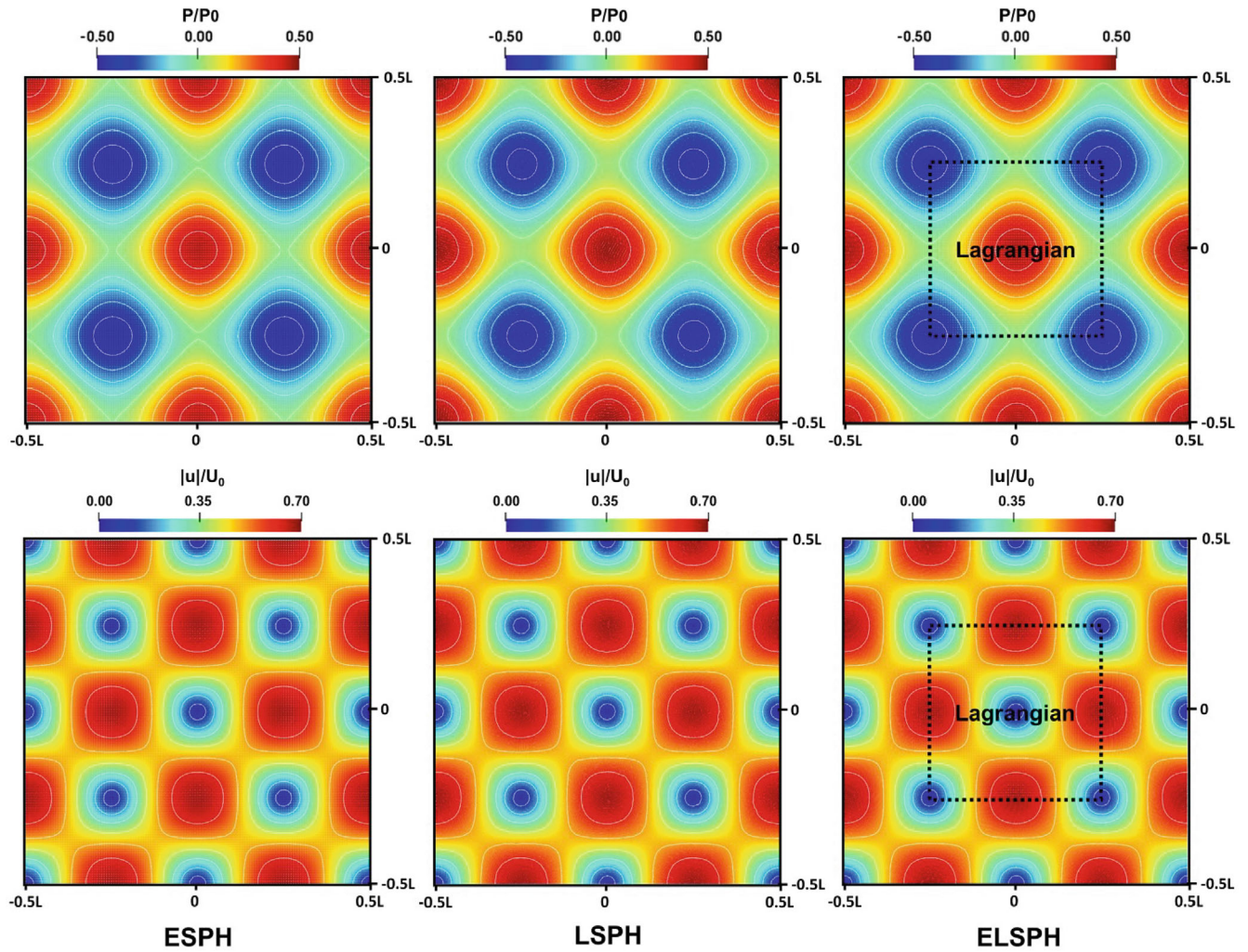


FIGURE 16 Taylor vortex: Velocity field contour (upper row) and pressure field contour (bottom row) at $t^* = 1$ ($L/\Delta X = 800$).

5.4 | 2D vortex shedding

In this section, the transient vortex shedding of a viscous flow past a cylinder is simulated at $Re = 200$. The flow passes through a bluff body, and vortices are generated at the rear end of the object. The vortices are regularly separated from each other and create a vortex structure, the so-called von Karman vortex street. In this simulation, the fluid domain is $[-4.5D, 4.5D] \times [-4D, 11D]$ and cylinder is placed on center $[0, 0]$, where the D is the diameter of the cylinder, and the cylinder is located at the origin. The speed of sound is set as $c_0 = 10U_{\max}$, where $U_{\max} = 1.5U$, U is the inlet velocity. In order to remove the effect of shockwave in the beginning stage, the fluid particles are accelerated up to desired inlet velocity as

$$\frac{du}{dt} = \begin{cases} 0.5U \frac{\pi}{t_0} \sin\left(\frac{\pi t}{t_0}\right) & t < t_0 \\ 0 & \text{otherwise} \end{cases}, \quad (59)$$

where desired time $t_0 = U/D$ in this study.³⁶ The upper and bottom wall are solved with slip boundary condition by not solving viscous force to remove the effect from the presence of walls.⁵¹

Simulations are performed using three different frameworks: ESPH, LSPH, and ELSPH. In the ELSPH, the Eulerian region is placed around the cylinder and is surrounded by Lagrangian particles, as depicted in Figure 19. Since the Eulerian region exist perpendicular to the flow direction, a transition zone is introduced to get stable results similar to Lind and Standby.¹⁷ The smooth transition zone is located back and forth along the flow direction. In this study, the parameter α

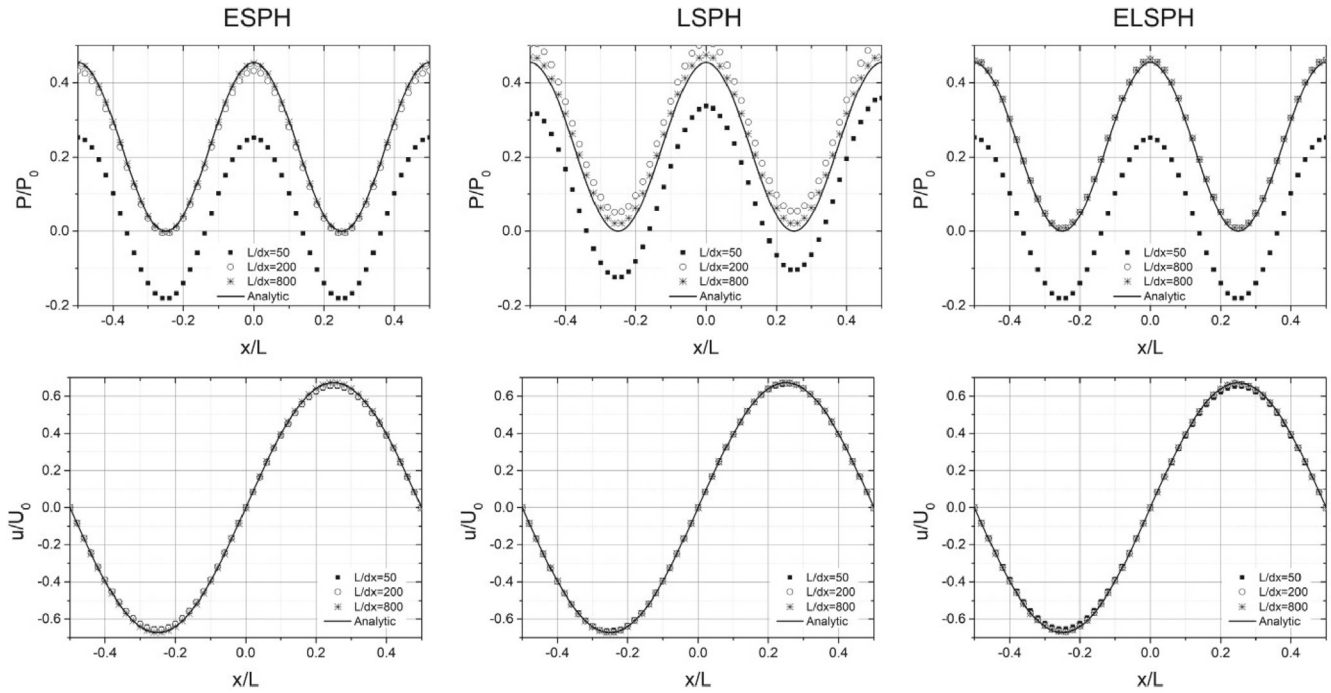


FIGURE 17 Taylor vortex: Pressure (upper row) and u-velocity distribution (bottom row) along the horizontal centerline ($y = 0$)

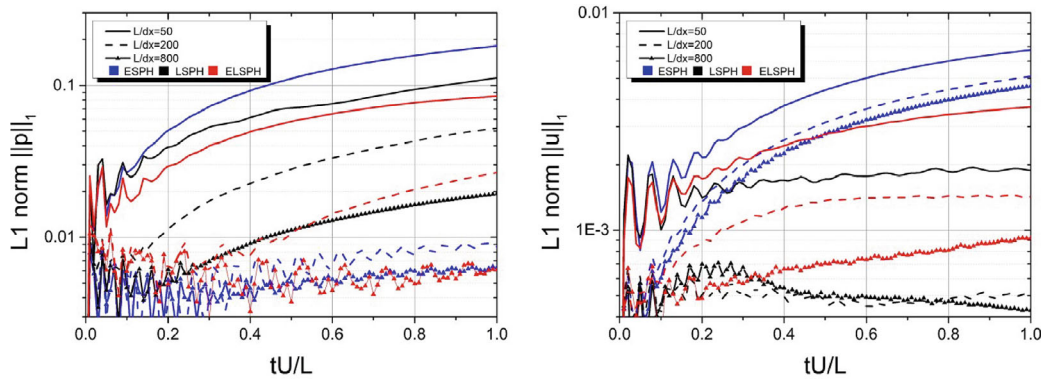


FIGURE 18 Taylor vortex: Temporal evolution of L1 error of pressure (left) and velocity (right) at $Re = 200$ for three methods.

is constructed with smooth sigmoid function.

$$\alpha = 0.5 \left[1 + \frac{\left(e^{\frac{(x-A+S)}{S}} - e^{\frac{(x-A-S)}{S}} \right)}{\left(e^{\frac{(x-A+S)}{S}} + e^{\frac{(x-A-S)}{S}} \right)} \right] \quad x \in [A, A + 2S] \quad (60)$$

Figure 19 illustrates the non-dimensional vorticity fields computed using the ESPH, LSPH, and ELSPH schemes at $t^* = 100$. The shaded area represents the transition region of the ELSPH scheme. Figure 20 demonstrates the time evolution of the force coefficients of the LSPH simulations at three particle resolutions ($D/dx = 20, 40, 80$). The force (drag and lift) coefficients are computed similarly to Marrone et al.,³⁷ as follows:

$$C_D = \frac{\sum_{j \in \text{solid}} f_x}{0.5 \rho v^2 D} \quad C_L = \frac{\sum_{j \in \text{solid}} f_y}{0.5 \rho v^2 D} \quad (61)$$

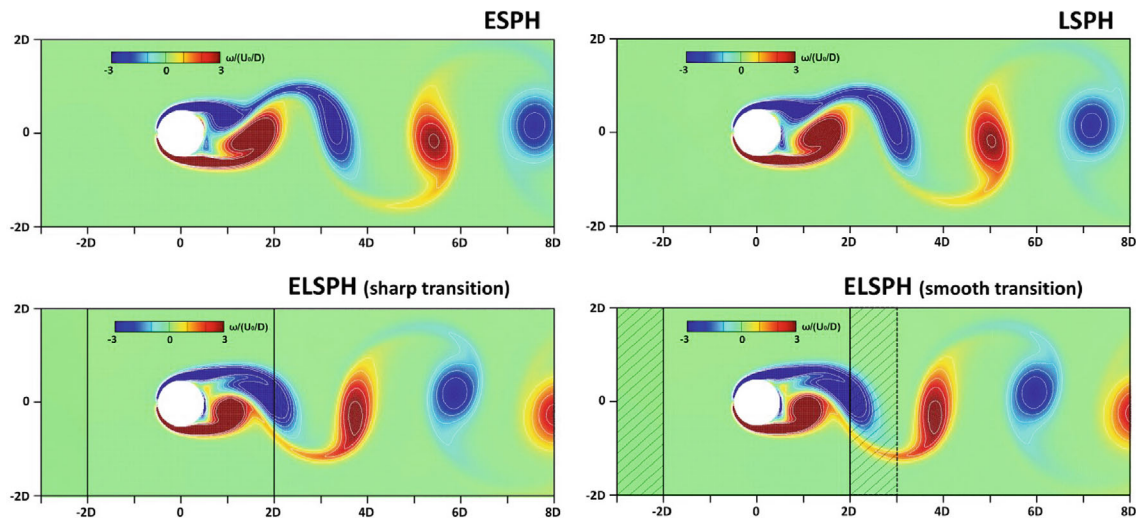


FIGURE 19 Vortex shedding: Vorticity contours of four methods at $t^* = 100$ for $D/dx = 80$

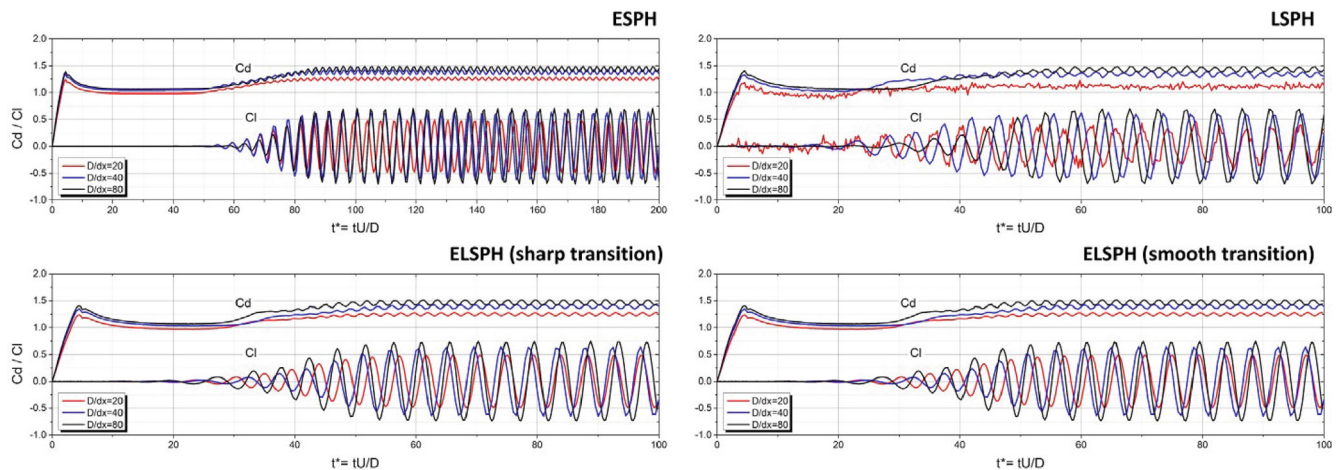


FIGURE 20 Vortex shedding: Force coefficients obtained using four method at three different resolutions ($D/\Delta x = 20, 40, 80$).

where C_D and C_L are the drag and lift coefficients, respectively, f_x and f_y denote the acceleration along the horizontal and perpendicular directions, and v and D denote the far-field velocity of the flow and diameter.

In the coarse spatial resolution ($D/dx = 20$), the particle space is insufficient to obtain the converged force coefficients. Converged results coinciding with previous studies are achieved in the cases with finer resolutions. The time histories of the drag coefficient obtained using the ELSPH scheme are depicted in Figure 20. In contrast to the LSPH cases, stable results are obtained with the coarse resolution ($D/dx = 20$) owing to uniform particle distribution near the bluff body. The force coefficients and Strouhal numbers ($St = fL/U$) obtained using the different schemes are compared with the reference solutions obtained using various numerical methods, as summarized in Table 3. Both the drag and lift coefficients obtained using the ESPH, LSPH, and ELSPH schemes are predicted similar to the reference results obtained using various schemes. In all cases, the phases differ slightly according to the resolution and numerical scheme. However, the Strouhal number (St) obtained from the averaged frequencies (f) are consistent with one another and the reference solutions.^{29,36–46}

5.5 | 2D sloshing in a rectangular tank

In this section, a benchmark simulation involving a free surface flow is conducted to validate the proposed method. The selected benchmark case is the harmonic sloshing tank problem. In this study, a body force is added to model

TABLE 3 Vortex shedding: Comparison of force coefficients on vortex shedding problem solved with different schemes ($Re = 200$).

	C_D	C_L	$St^\#$
ESPH ($D/dx = 20$)	1.255 ± 0.03	± 0.478	0.226
ESPH ($D/dx = 40$)	1.379 ± 0.05	± 0.623	0.218
ESPH ($D/dx = 80$)	1.438 ± 0.05	± 0.692	0.215
LSPH ($D/dx = 20$)	1.111 ± 0.08	± 0.184	-
LSPH ($D/dx = 40$)	1.367 ± 0.08	± 0.624	0.220
LSPH ($D/dx = 80$)	1.431 ± 0.06	± 0.695	0.215
ELSPH ^a ($D/dx = 20$)	1.251 ± 0.03	± 0.489	0.224
ELSPH ^a ($D/dx = 40$)	1.381 ± 0.05	± 0.638	0.217
ELSPH ^a ($D/dx = 80$)	1.447 ± 0.05	± 0.714	0.215
ELSPH ^b ($D/dx = 20$)	1.252 ± 0.03	± 0.490	0.227
ELSPH ^b ($D/dx = 40$)	1.389 ± 0.05	± 0.634	0.219
ELSPH ^b ($D/dx = 80$)	1.461 ± 0.05	± 0.717	0.215
Marrone et al. ³⁷ (SPH)	1.35 ± 0.05	± 0.68	0.200
Sun et al. ²⁹ (SPH)	1.35 ± 0.05	± 0.73	-
Tafuni et al. ⁵² (SPH)	1.46	± 0.693	0.206
Negi et al. ³⁸ (SPH)	1.524 ± 0.05	± 0.722	0.210
Zhang et al. ³⁶ (SPH)	1.571 ± 0.05	± 0.81	0.23
Rossi et al. ³⁹ (DVH)	1.35 ± 0.05	± 0.68	
Belov et al. ⁴⁰ (FVM)	1.19 ± 0.042	± 0.64	0.192
Braza et al. ⁴¹ (FVM)	1.40 ± 0.05	± 0.75	-
Choi et al. ⁴² (FVM)	1.36 ± 0.048	± 0.64	0.191
Ng et al. ⁴³ (FVM)	1.375 ± 0.05	± 0.724	
Rosenfield et al. ⁴⁴ (FVM)	1.31 ± 0.04	± 0.64	0.20
Lecointe and Piquet. ⁴⁵ (FDM) (2nd/4th order)	1.46 ± 0.04	± 0.70	0.227
	1.58 ± 0.035	± 0.50	0.19
Rogers and Kwak ⁴⁶ (FDM) (3rd/5th order)	1.29 ± 0.05	± 0.75	0.16
	1.23 ± 0.05	± 0.65	0.185

^aWithout transition zone.^bWith transition zone.

the periodic sloshing motion in the tank. The OPS technique explained in Section 2.3 is applied to regularize the particle distribution near the Eulerian–Lagrangian transition region and the free surface. The dimensions of the fluid domain in the tank are $[D \times H]$, and the motion of the tank is represented using the first-order linear wave theory, as follows:

$$x(t) = A \cos(\omega_n t)$$

$$\omega_n^2 = |g|k_n \tanh(k_n H) \quad (62)$$

where A is the amplitude, ω_n is the angular frequency, g is the gravitational acceleration, k_n is the wave number, H is the initial wave depth inside the tank, and n is the free surface mode. The amplitude and frequency are $0.03D$ and ω_1 , as in.⁴⁷ Same with their study, the initial The LSPH and ELSPH results are compared to the finite element method (FEM) solution⁴⁷ in Figure 21.

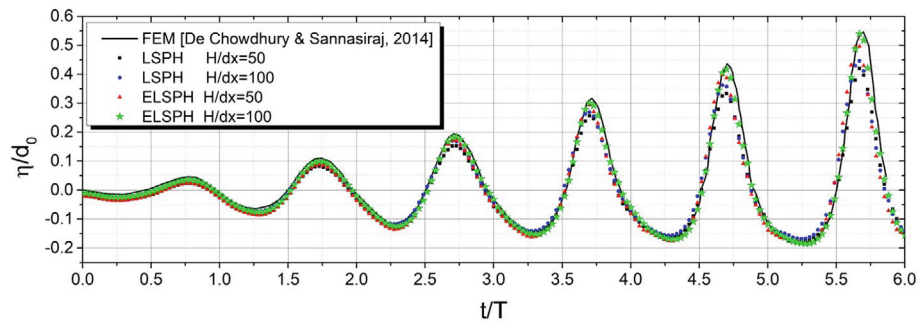


FIGURE 21 Sloshing: The comparison on wave height variation of the left wall in time for all cases ($H/\Delta x = 50, 100$).

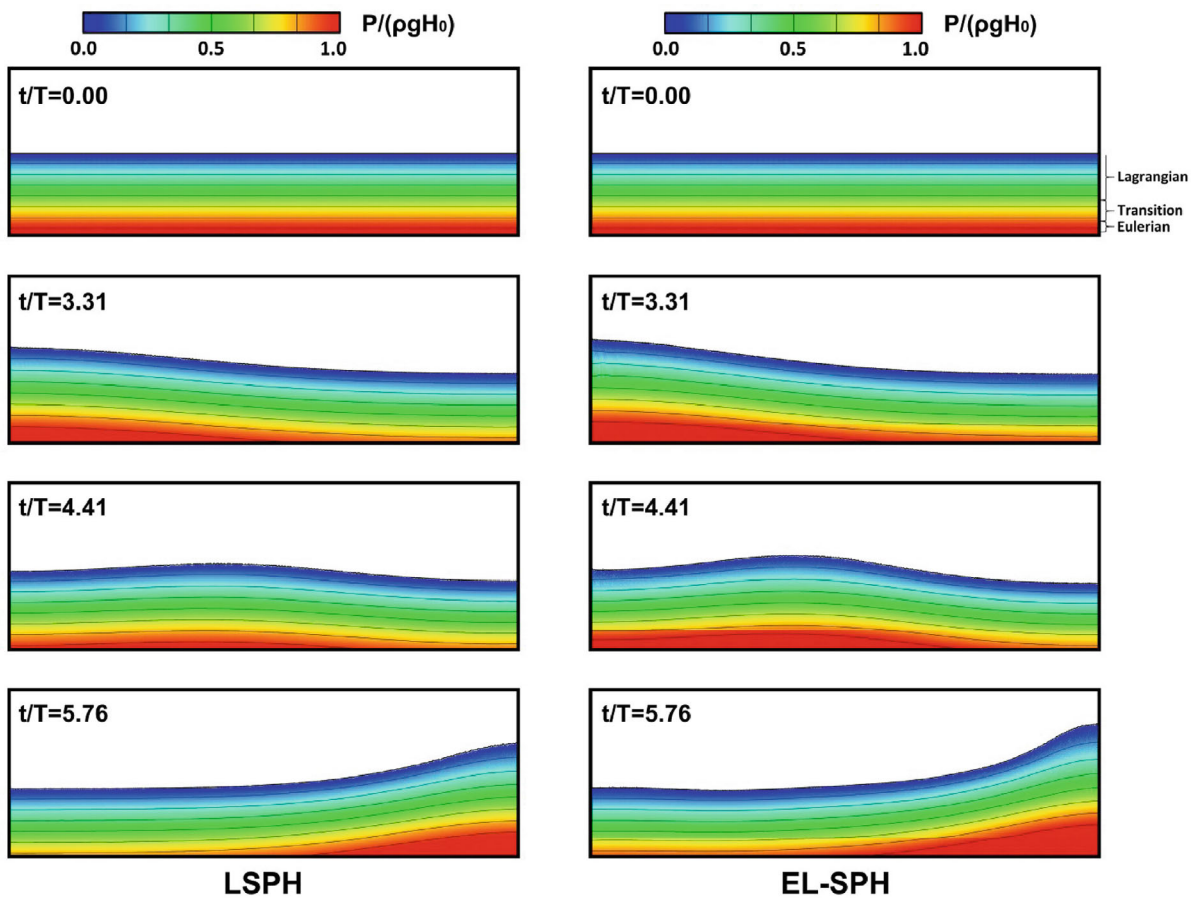


FIGURE 22 Sloshing: Pressure contour of LSPH (left) and ELSPH (right) at a fine resolution ($H/\Delta x = 100$).

In the case of ELSPH, the Eulerian region is set from the bottom of the tank to the quarter-point: one quarter above is set as the transition region, and the other area is set as the Lagrangian region. Two different spatial resolutions ($H/dx = 50, 100$) are selected, and both frameworks yielded accurate predictions of wave elevation on the left wall where the H is the initial height of fluid domain. The ELSPH scheme predicted the wave height slightly more accurately than the LSPH scheme due to the characteristics of WCSPP—some compression of particle volume is enabled in the Lagrangian framework, whereas uniform particle distribution is maintained in the Eulerian region of the Eulerian–Lagrangian framework. In this respect, the speed of sound was adjusted to $c_0 = 160\sqrt{gd}$, but a slightly lower height was obtained compared to the FEM result. Similar trends were reported by De Chowdhury and Sannasiraj.⁴⁷ Figure 22 illustrates the pressure fields of the two different schemes at a fine resolution ($H/dx = 100$).

5.6 | 2D vortex shedding on free surface

The vortex shedding problem on a free surface was considered to demonstrate the accuracy and flexibility of the proposed scheme. The vortex shedding problem on a free surface has been solved in only a few studies to the best of our knowledge. The detailed SPH simulation of Boucasse et al.⁴⁸ is referenced in this study, which performed a simulation that involved a detailed parametric study and an ultrafine resolution ($D/dx = 100$) with large smoothing resolution $h = 5.8dx$ where d is the diameter of cylinder. The cylinder was submerged in water, and it was assumed that $Re = 180$. The Froude number (Fr) in this problem was defined as follows:

$$Fr_d = \frac{U}{\sqrt{gd}} \quad (63)$$

where U is the flow velocity. Calculations were performed for the case where $Fr_d = 1.0$ and $h/d = 2.5$ with the submerged depth of h . The speed of sound is set as $c_0 = \max(10U_{\max}, 20\sqrt{gH})$, where $U_{\max} = 1.5U$, where U is the inlet velocity. The upper and bottom wall are solved with slip boundary condition by not solving viscous force to remove the effect from the presence of walls (adami).

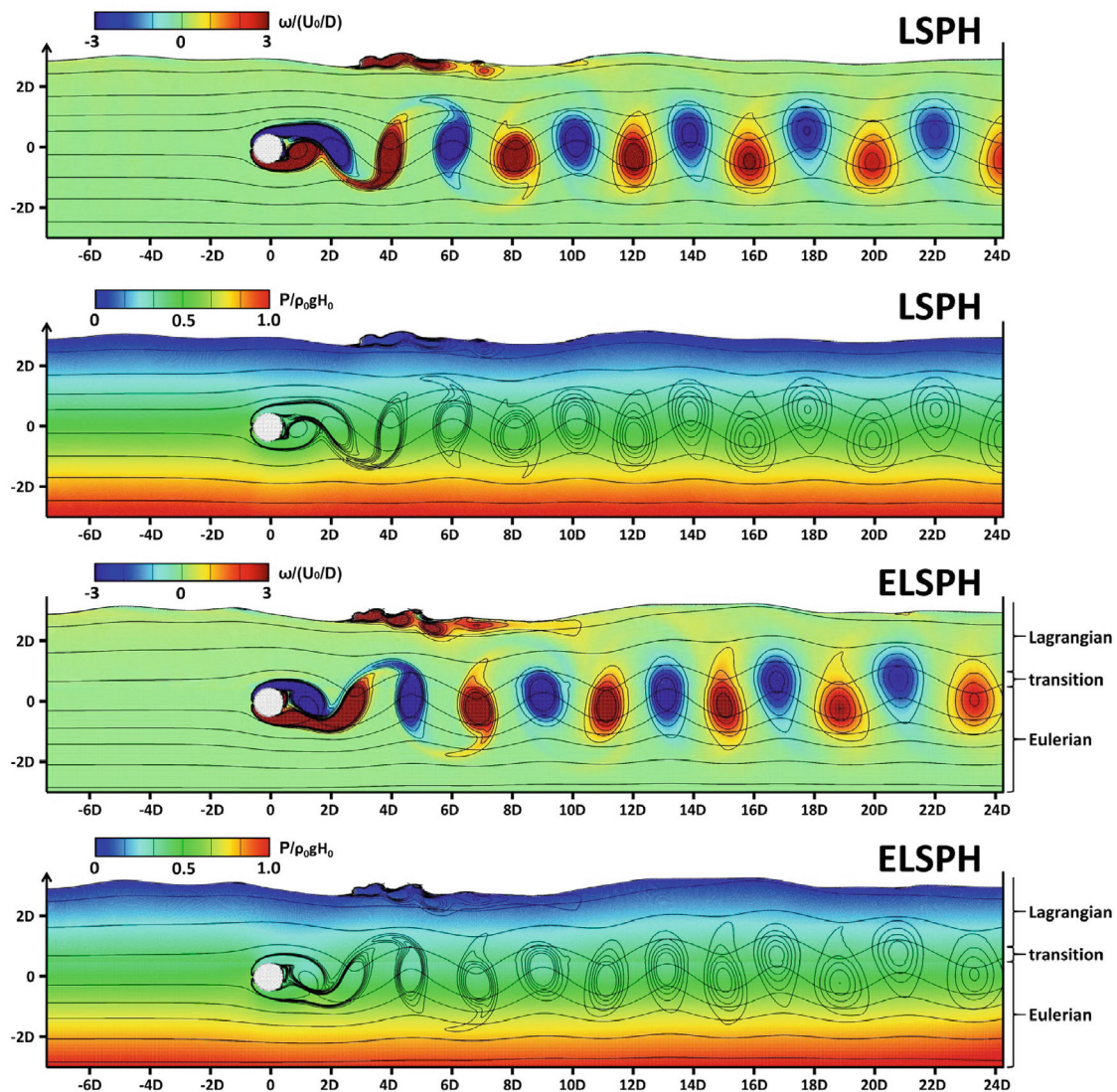


FIGURE 23 Vertex shedding on free surface: Vorticity and pressure contours at $t^* = 60$ for $D/\Delta x = 40$.

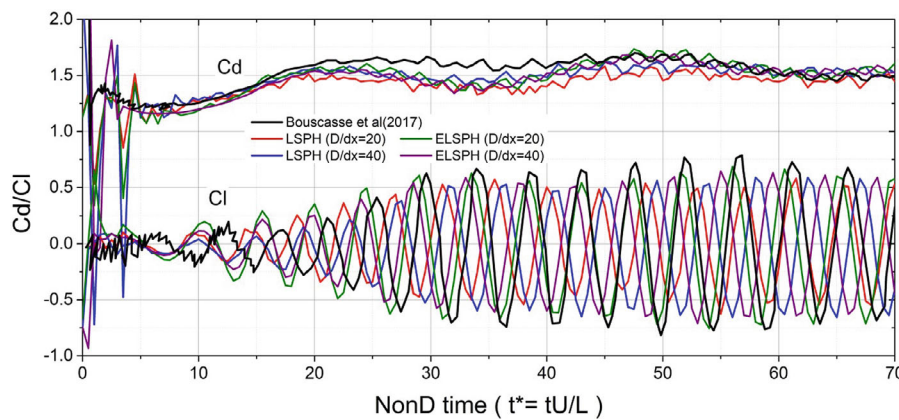


FIGURE 24 Vertex shedding on free surface: Force coefficients calculated using LSPH ($D/\Delta x = 20, 40$) and ELSPH ($D/\Delta x = 20, 40$).

The simulation was performed using the LSPH and ELSPH schemes. In the case of the ELSPH case, the Eulerian region was set from the bottom to top of the cylindrical object, the transition region $D/2$ above it, and the Lagrangian region above it. The depth of the fluid domain was determined at the point where the bottom surface had a negligible influence on the flow, as in Reference 48. The vorticity plots at the end of simulations are depicted along with the isovorticity line in Figure 23—the shapes of the isovorticity lines illustrate the similarities between the two frameworks.

The force coefficients are plotted in Figure 24, along with the reference results. They periodically rise and fall under the influence of the free surface, in contrast to the case without the free surface, as in Section 3.4. The tendencies observed in the reference results were accurately predicted by both schemes. In the LSPH case, the trend was more similar to that of the ELSPH case and the reference study at a finer resolution ($D/dx = 40$).

5.7 | 2D natural convection in a square cavity

The natural convection in a square cavity, which include convective heat transfer is solved as a final benchmark. In this problem, a flow is driven in a fully confined square cavity by the convective heat transfer. The Prandtl number ($Pr = \mu c_p / k$) is maintained to 0.71 and four values of the Rayleigh number ($Ra = Gr \cdot Pr$) are tested, namely 10^3 , 10^4 , 10^5 , and 10^6 . The Grashof number (Gr) is defined as $Gr = \beta g (T - T_0) L^3 / \nu^2$, where ν is the kinematic viscosity, defined as $\nu = \mu / \rho$ and T_0 is the reference temperature. The left and right walls are in isothermal condition, the cold wall being the right, and the lower and upper walls are adiabatic. The no-slip boundary condition and pressure Neumann boundary condition is enforced on all sided-walls. Similar to the lid-driven problem, simulations are conducted using the ESPH, LSPH, and ELSPH schemes with same region division. The fluid domain size is $[L \times L]$ and the discretization of fluid particles is $L/dx = 400$ for all simulations. The speed of sound is set as $c_0 = 10U_{\max}$, where $U_{\max} = \beta g (T_H - T_L) L$, where T_H is the temperature of the hot wall and T_L is the temperature of the cold wall (Figure 25).

The converged results of the dimensionless temperature ($T^* = (T - T_L) / (T_H - T_L)$) and velocity ($v^* = vL / \alpha_D$) fields for $Ra = 10^5$ and 10^6 are shown in Figure 25, where α_D is the thermal diffusivity, defined as $\alpha_D = k c_p / \rho$. Close agreement is shown between the three different schemes, and the field variables are free of noise across the transition region. In Figure 26, the vertical velocity across the horizontal centerline are compared with discrete singular convolution (DSC) results and shows very good agreement.⁴⁹ Figure 27 shows the distribution of the local Nusselt number ($Nu = -\partial T^* / \partial y^*$) along the hot wall and temperature profile across the horizontal centerline for the four values of the Rayleigh number with three different schemes. Both results are compared with DSC results. The global trend of the local Nusselt number is well predicted, but slight discrepancies are shown in the peak of the curve at $Ra = 10^5$ and 10^6 . However, the temperature profiles are in very good agreement with DSC results near the hot wall. The similar results are reported by Leroy et al.⁵⁰ and they concluded that the discrepancies might be induced by the lack of accuracy in gradient operator, which is used to compute the Nusselt number.

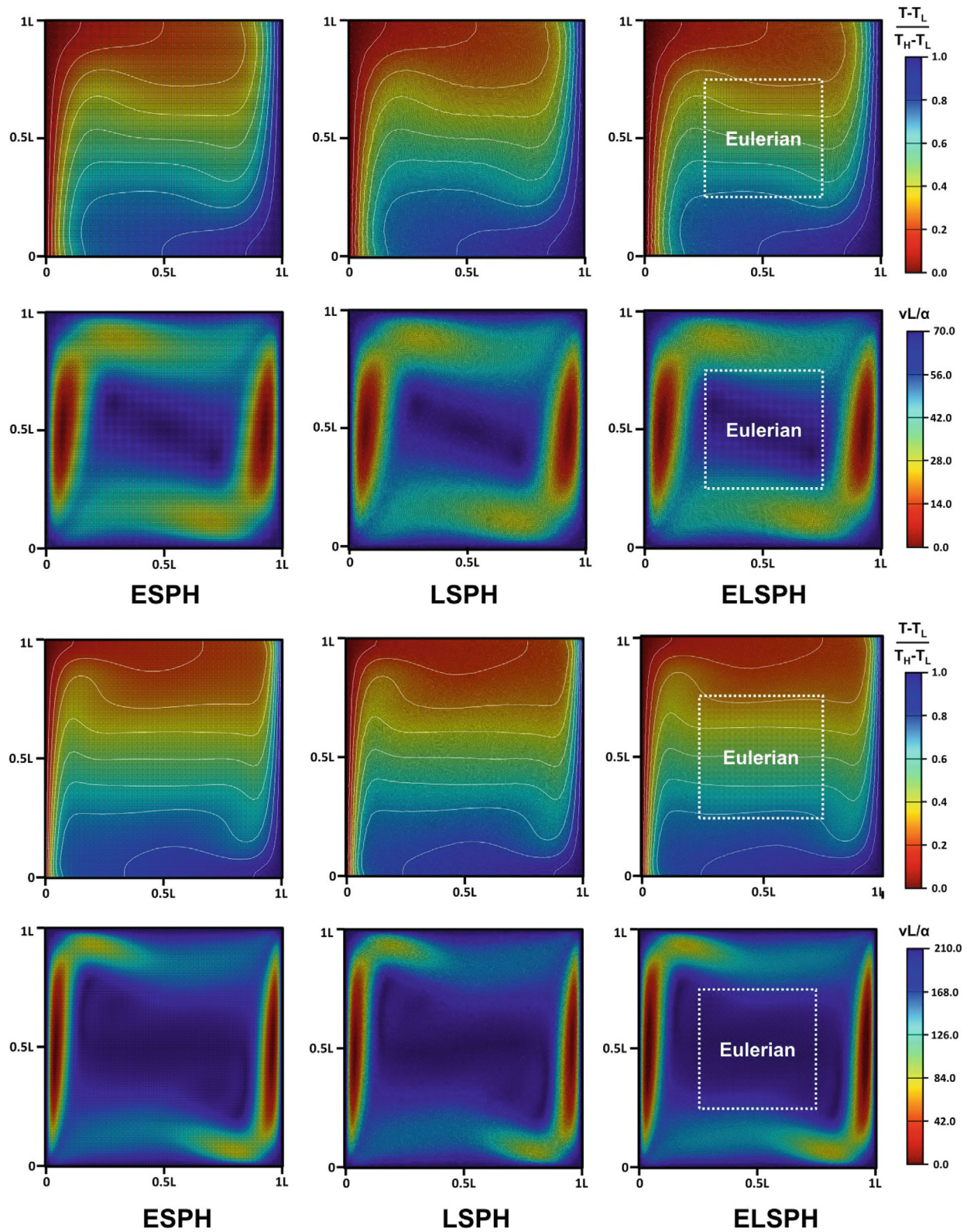


FIGURE 25 Temperature and velocity fields obtained using different schemes at $Ra = 10^5$ (upper two rows) and $Ra = 10^6$ (lower two rows)

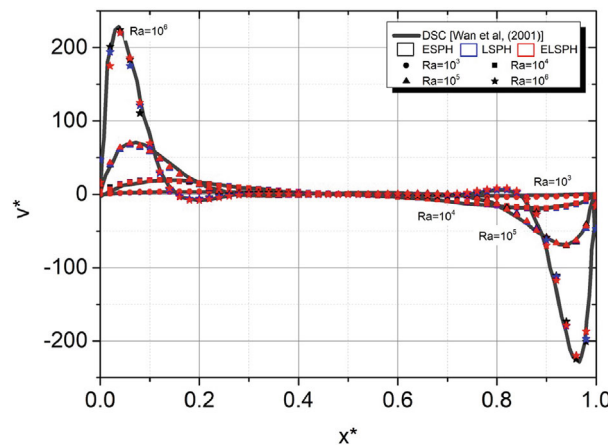


FIGURE 26 Natural convection: Vertical velocity profile along horizontal centerline at $Ra = 10^3$ to $Ra = 10^6$

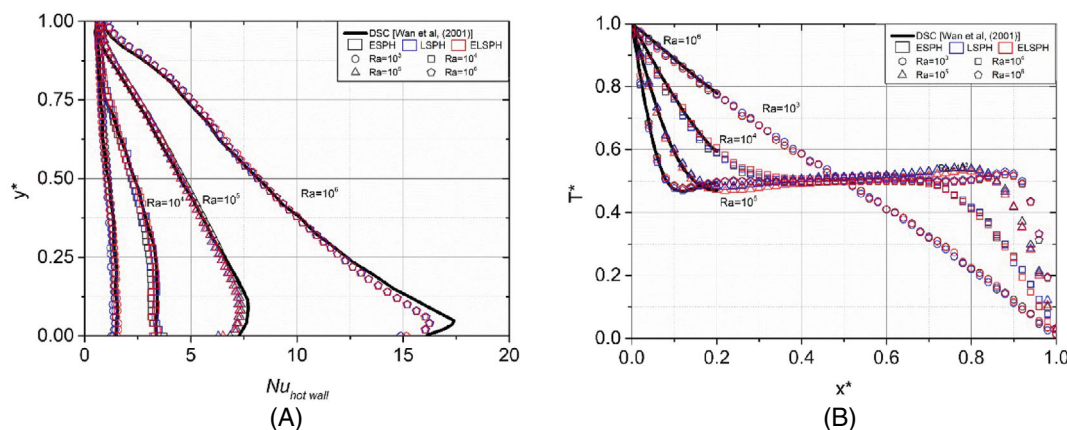


FIGURE 27 Natural convection: (A) Temperature distribution along the horizontal centerline near the hot wall and (B) evolution of the Nusselt number at the hot wall at $Ra = 10^3$ to $Ra = 10^6$

6 | CONCLUSIONS

The simple and novel ELSPH scheme was proposed to exploit both Eulerian and Lagrangian schemes in the WSPH context. The governing equations were written in the ALE formalism with an arbitrary transport velocity, and the arbitrary transport velocity was adjusted to separate the Eulerian and Lagrangian regions. The anisotropic particle structure and penetration between regions was prevented by using the stabilizing ability of PST. The OPS algorithm was applied to ensure a uniform particle distribution near the free surface. The simple and straightforward mass correction method was proposed to consider the variation of mass and volume due to the weakly compressible assumption. Through the complement of mass correction and particle disorder correction, the improvement on particle unity condition is acquired regardless of the arbitrary transport velocity as validated with 1D isentropic shock tube problem which involve high compressibility.

Two steady-state test cases were considered for testing the ELSPH scheme: pipe flow at $Re = 80$ and lid-driven cavity at $Re = 1000$. The first benchmark test demonstrated that the ESPH, LSPH, and ELSPH schemes provided the correct velocity and pressure profiles relative to the analytical solution. The accurate velocity distribution and stable pressure distribution could be achieved by setting the fluid particles near the wall as Eulerian region. The results of the second benchmark demonstrated the accuracy and consistency of the ELSPH scheme, with the close agreement between the three different schemes and the FDM results. The three methods (ESPH, LSPH, and ELSPH) showed good convergence and LSPH predicted the local velocity slightly better than ESPH and ELSPH. However, ESPH and ELSPH predicted the position and vorticity of secondary vortices more accurately than LSPH.

Two transient benchmark cases were studied to test the convergence of the ELSPH scheme. The scheme was first tested by applying it to the Taylor–Green vortices problem. Three schemes predicted the local velocity and pressure distribution with good convergence to the analytical results. The global error of pressure and velocity are evaluated with the time evolution of L1 norm error and shown to be convergent for three methods. The ESPH and ELSPH with Eulerian particles around the periodic boundaries predicted more accurate pressure distribution. The velocity distribution is slightly better represented using LSPH but the error of all three schemes were very small. Next, the vortex shedding problem at $Re = 200$ was considered. In this problem, three schemes are used and the transition zone is adopted to get stable results. All three schemes yielded convergent results similar with other numerical methods. The LSPH required a higher resolution to get the stable force coefficients than the pure Eulerian scheme. The stable force coefficients are reproduced by placing the Eulerian particles around the solid bluff body at a lower spatial resolution than the LSPH scheme.

Then, the sloshing problem was studied to extend the ELSPH scheme to a case involving a free surface. The Eulerian particles were restricted to the inner region and replaced with the Lagrangian particles near the free surface. The wave height was predicted slightly insufficiently when the Lagrangian scheme was used due to the weakly compressible assumption on particle volumes. More accurate predictions can be achieved by using the Eulerian–Lagrangian scheme due to the uniform particle distribution. The vortex shedding on the free surface was solved as a next-step benchmark for more complicated free surface flows. Both the Lagrangian and Eulerian–Lagrangian schemes yielded force coefficients similar to the reference results. The results obtained using the Eulerian–Lagrangian scheme were closer to the reference results at lower resolutions.

Finally, the ELSPH scheme was extended to handle the convective heat transfer for thermal–hydraulic problems. The natural convection in a square cavity in various Rayleigh numbers are considered as a last benchmark and consistent results are obtained between all the three different schemes. Slight under-prediction are shown in the local Nu number due to lack of accuracy of the gradient operator when computing Nu . However, the global tendency of Nu number near the hot wall, temperature and velocity are well converged and matched with DSC results.

In conclusion, the accuracy, consistency, and convergence of the ELSPH scheme was demonstrated using several benchmarks. It was shown that the unity condition is satisfied by the complement of straightforward mass correction and disorder correction. Also, more stable and accurate results could be achieved by placing Eulerian region near the wall or rigid boundary. The overall investigation results highlight the applicability and flexibility of the proposed scheme from various hydrodynamics problems to thermal–hydraulics problems. In the future, the ELSPH scheme will be extended to handle multi-resolution and multi-physics problems, including phase change and fluid–solid interactions.

ACKNOWLEDGMENTS

This work was supported by the National Research Foundation of Korea (NRF) Grant funded by the Korea government (MSIT) (Nos. 2019R1F1A1060159 and 2021M2D2A1A03046881).

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Hee Sang Yoo  <https://orcid.org/0000-0002-5497-6725>

REFERENCES

1. Gingold RA, Monaghan JJ. Smoothed particle hydrodynamics: theory and application to non-spherical stars. *Monthly Not Royal Astron Soc.* 1977;181(3):375–389.
2. Lucy LB. A numerical approach to the testing of the fission hypothesis. *Astron J.* 1977;82:1013–1024.
3. Jo YB, Park SH, Choi HY, Jung HW, Kim YJ, Kim ES. SOPHIA: development of Lagrangian-based CFD code for nuclear thermal-hydraulics and safety applications. *Ann Nucl Energy.* 2019;1(124):132–149.
4. Ye T, Pan D, Huang C, Liu M. Smoothed particle hydrodynamics (SPH) for complex fluid flows: recent developments in methodology and applications. *Phys Fluids.* 2019;31(1):011301.
5. Zhang AM, Sun PN, Ming FR, Colagrossi A. Smoothed particle hydrodynamics and its applications in fluid–Structure interactions. *J Hydrodyn Ser B.* 2017;29(2):187–216.
6. Lluch È, De Craene M, Bijmens B, et al. Breaking the state of the heart: meshless model for cardiac mechanics. *Biomech Model Mechanobiol.* 2019;18(6):1549–1561.

7. Cleary PW, Harrison SM, Sinnott MD, et al. Application of SPH to single and multiphase geophysical, biophysical and industrial fluid flows. *Int J Comput Fluid Dyn.* 2021;35(1–2):22–78.
8. Monaghan JJ. Simulating free surface flows with SPH. *J Comput Phys.* 1994;110(2):399–406.
9. Molteni D, Colagrossi A. A simple procedure to improve the pressure evaluation in hydrodynamic context using the SPH. *Comput Phys Commun.* 2009;180(6):861–872.
10. Antuono M, Colagrossi A, Marrone S, Molteni D. Free-surface flows solved by means of SPH schemes with numerical diffusive terms. *Comput Phys Commun.* 2010;181(3):532–549.
11. Zhang C, Hu XY, Adams NA. A weakly compressible SPH method based on a low-dissipation Riemann solver. *J Comput Phys.* 2017;335:605–620.
12. Oger G, Marrone S, Le Touzé D, De Leffe M. SPH accuracy improvement through the combination of a quasi-Lagrangian shifting transport velocity and consistent ALE formalisms. *J Comput Phys.* 2016;313:76–98.
13. Vila JP. On particle weighted methods and smooth particle hydrodynamics. *Math Models Methods Appl Sci.* 1999;9(02):161–209.
14. Collé A, Limido J, Vila J-P. An accurate multi-regime SPH scheme for barotropic flows. *J Comput Phys.* 2019;388:561–600.
15. Collé A, Limido J, Unfer T, Vila JP. An accurate SPH scheme for hypervelocity impact modeling. Proceedings of the Hypervelocity Impact Symposium 2019; Vol. 883556, 2019;V001T02A005; American Society of Mechanical Engineers.
16. Antuono M, Sun PN, Marrone S, Colagrossi A. The δ -ALE-SPH model: an arbitrary Lagrangian-Eulerian framework for the δ -SPH model with particle shifting technique. *Comput Fluids.* 2021;2(16):104806.
17. Lind SJ, Stansby PK. High-order Eulerian incompressible smoothed particle hydrodynamics with transition to Lagrangian free-surface motion. *J Comput Phys.* 2016;326:290–311.
18. Fourtakas G, Stansby PK, Rogers BD, Lind SJ. An Eulerian–Lagrangian incompressible SPH formulation ELI-SPH connected with a sharp interface. *Comput Methods Appl Mech Eng.* 2018;329:532–552.
19. Wendland H. Piecewise polynomial, positive definite and compactly supported radial functions of minimal degree. *Adv Comput Math.* 1995;4(1):389–396.
20. Liu MB, Liu GR. Smoothed particle hydrodynamics (SPH): an overview and recent developments. *Arch Comput Methods Eng.* 2010;17(1):25–76.
21. Colagrossi A, Antuono M, Souto-Iglesias A, le Touzé D. Theoretical analysis and numerical verification of the consistency of viscous smoothed-particle-hydrodynamics formulations in simulating free-surface flows. *Phys Rev E.* 2011;84(2):026705.
22. Hopkins PF. A general class of Lagrangian smoothed particle hydrodynamics methods and implications for fluid mixing problems. *Monthly Not Royal Astron Soc.* 2013;428(4):2840–2856.
23. Saitoh TR, Makino J. A density-independent formulation of smoothed particle hydrodynamics. *Astrophys J.* 2013;768(1):44.
24. Cabezón RM, García-Senz D, Figueira J. SPHYNX: an accurate density-based SPH method for astrophysical applications. *Astron Astrophys.* 2017;606:A78.
25. García-Senz D, Wissing R, Cabezón RM. Axisymmetric magneto-hydrodynamics with SPH. arXiv Preprint arXiv:2206.05324, 2022.
26. Xu R, Peter S, Dominique L. Accuracy and stability in incompressible SPH ISPH based on the projection method and a new approach. *J Comput Phys.* 2009;228(18):6703–6725.
27. Lind SJ, Xu R, Stansby PK, Rogers BD. Incompressible smoothed particle hydrodynamics for free-surface flows: a generalised diffusion-based algorithm for stability and validations for impulsive flows and propagating waves. *J Comput Phys.* 2012;231(4):1499–1523.
28. Khayyer A, Hitoshi G, Yuma S. Comparative study on accuracy and conservation properties of two particle regularization schemes and proposal of an optimized particle shifting scheme in ISPH context. *J Comput Phys.* 2017;3(32):236–256.
29. Sun PN, Colagrossi A, Marrone S, Zhang AM. The δ plus-SPH model: simple procedures for a further improvement of the SPH scheme. *Comput Methods Appl Mech Eng.* 2017;3(15):25–49.
30. Marongiu JC. *Lagrangian Numerical Method for the Simulation of Free Surface Flows: Application to Pelton Turbines*. PhD Thesis. Ecully, Central School of Lyon; 2007.
31. Xia X, Liang Q. A GPU-accelerated smoothed particle hydrodynamics (SPH) model for the shallow water equations. *Env Model Softw.* 2016;75:28–43.
32. White FM. *Fluid Mechanics*. Tata McGraw-Hill Education; 1979.
33. Ghia UK, Ghia KN, Shin CT. High-resolutions for incompressible flow using the Navier-Stokes equations and a multigrid method. *J Comput Phys.* 1982;48(3):387–411.
34. Lee ES, Moulinec C, Xu R, Violeau D, Laurence D, Stansby P. Comparisons of weakly compressible and truly incompressible algorithms for the SPH mesh free particle method. *J Comput Phys.* 2008;227(18):8417–8436.
35. Taylor GI, Green AE. Mechanism of the production of small eddies from large ones. *Proc Royal Soc Lond Ser A-Math Phys Sci.* 1937;158(895):499–521.
36. Zhang S, Zhang W, Zhang C, Hu X. A Lagrangian free-stream boundary condition for weakly compressible smoothed particle hydrodynamics. arXiv Preprint arXiv:2206.06875, 2022.
37. Marrone S, Colagrossi A, Antuono M, Colicchio G, Graziani G. An accurate SPH modeling of viscous flows around bodies at low and moderate Reynolds numbers. *J Comput Phys.* 2013;245(452):456–475.
38. Negi P, Ramachandran P, Haftu A. An improved non-reflecting outlet boundary condition for weakly-compressible SPH. *Comput Methods Appl Mech Eng.* 2020;367:113–119.
39. Rossi E, Colagrossi A, Bouscasse B, Graziani G. The diffused vortex hydrodynamics method. *Commun Comput Phys.* 2015;18(2):351–379.

40. Belov A, Martinelli L, Jameson A. A new implicit algorithm with multigrid for unsteady incompressible flow calculations. Proceedings of the 33rd Aerospace Sciences Meeting and Exhibit; 1995:49.
41. Braza M, Chassaing PH, Minh HH. Numerical study and physical analysis of the pressure and velocity fields in the near wake of a circular cylinder. *J Fluid Mech*. 1986;165:79-130.
42. Rogers SE, Kwak D. Upwind differencing scheme for the time-accurate incompressible Navier–Stokes equations. *AIAA J*. 1990;28(2):253-262.
43. Choi JJ, Oberoi RC, Edwards JR, Rosati JA. An immersed boundary method for complex incompressible flows. *J Comput Phys*. 2007;224(2):757-784.
44. Ng YT, Min C, Gibou F. An efficient fluid–Solid coupling algorithm for single-phase flows. *J Comput Phys*. 2009;23(228):8807-8829.
45. Lecointe Y, Piquet J. On the use of several compact methods for the study of unsteady incompressible viscous flow round a circular cylinder. *Comput Fluids*. 1984;12(4):255-280.
46. Rosenfeld M, Kwak D, Vinokur M. A fractional step solution method for the unsteady incompressible Navier–Stokes equations in generalized coordinate systems. *J Comput Phys*. 1991;94(1):102-137.
47. De Chowdhury S, Sannasiraj SA. Numerical simulation of 2D sloshing waves using SPH with diffusive terms. *Applied Ocean Research*. 2014;47:219-240.
48. Bouscasse B, Colagrossi A, Marrone S, Souto-Iglesias A. SPH modelling of viscous flow past a circular cylinder interacting with a free surface. *Comput Fluids*. 2017;1(46):190-212.
49. Wan DC, Patnaik BSV, Wei GW. A new benchmark quality solution for the buoyancy-driven cavity by discrete singular convolution. *Numer Heat Transf B Fund*. 2001;40(3):199-228.
50. Leroy A, Violeau D, Ferrand M, Joly A. Buoyancy modelling with incompressible SPH for laminar and turbulent flows. *Int J Numer Methods Fluids*. 2015;78(8):455-474.
51. Adami S, Hu XY, Adams NA. A generalized wall boundary condition for smoothed particle hydrodynamics. *J Comput Phys*. 2012;231(21):7057-7075.
52. Tafuni A, Domínguez JM, Vacondio R, Crespo AJ. A versatile algorithm for the treatment of open boundary conditions in smoothed particle hydrodynamics GPU models. *Comput Methods Appl Mech Eng*. 2018;3(42):604-624.

How to cite this article: Yoo HS, Jo YB, Kim JW, Kim ES, Choi TS. A simple Eulerian–Lagrangian weakly compressible smoothed particle hydrodynamics method for fluid flow and heat transfer. *Int J Numer Methods Eng*. 2023;124(4):928-958. doi: 10.1002/nme.7148